

INDEPENDENT REPLICATION

Intraday momentum on SPY

An independent replication and cost analysis of blackswan-
quants/intraday-momentum

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§0. Abstract

blackswan-quants/intraday-momentum implements five variants of an intraday momentum strategy on SPY and presents them as a progression that culminates in Strategy 4, which the paper describes as Pareto-dominant. This report re-runs the five published backtests, checks them against the authors' own result files, adds fifty-five more, and reaches four conclusions.

First, there is a signal. A benchmark with no signal at all, on the same instrument, with the same leverage rule and the same trading window, loses 36.1% over the period in which the strategy gains 271.4%, and it loses in both directions: 30.7% with the position reversed.

Aligning the benchmark's entry time with the strategy's own removes a further 17.8 points from the benchmark: the opening half-hour, which the thirty-minute cadence excludes by construction, is worth that much to a passive leveraged position in this sample. The edge is more sensitive to execution costs than its size suggests, however. Reconstructing the effect of transaction costs in closed form from a single backtest, the 271.4% falls to zero at 2.2 basis points of slippage per trade beyond the quoted spread, while leveraged buy-and-hold at the same leverage survives up to 218.

Second, the comparison on raw return is the less informative one. The strategy's 271% sits below the 350% of leveraged buy-and-hold, but the gap is about the size of the margin interest that the backtest does not charge, so it does not support an argument in either direction. The rest of the comparison does hold: a beta of -0.05 and a quarter of the drawdown, 9.0% vs. 40.2%. Both come from the signal, not from the schedule: the same leveraged position held without a signal, and flat every night as well, has a beta of 0.80 and a drawdown of 40.5%. What being flat every night explains is the missing equity premium, which over this window accrues almost entirely overnight.

Third, the two mechanisms that Strategy 4 combines are not independent, as the paper's Hypothesis 1 assumes: they are partial substitutes. Separating the steps of the ladder, which the repository's own Strategy 1 allows, shows that the five-minute exit check costs return and the persistence gate more than recovers it; but the value of that exit package falls by about 0.05 in log wealth when the EMA entry filter is present, under both exit thresholds. Strategy 4 still dominates Strategies 2 and 3, as the paper reports; the reason it gives does not hold.

Fourth, the best of the sixteen configurations examined is not in the repository, but it is one line away from Strategy 4 and its lead is small. Adding to Strategy 4 an entry condition that is present in the code of Strategy 2 but absent from both published descriptions gives S4', which closes with 3.4% more capital over eight years, about 0.4% per year, on the strength of the 22 to 26 trades the condition blocks, at the price of a slightly deeper drawdown; whether that lead survives outside this sample is not tested here. Corrected for selection across sixteen

trials with the Deflated Sharpe Ratio, S4' remains distinguishable from search noise, although the test is against a null of zero edge, not against Strategy 4. That check matters because the one significance statistic the paper reports, QuantConnect's PSR, has changed definition on the platform since the authors' runs, and the same code now returns about forty points less.

§1. The repository and the paper

`blackswan-quants/intraday-momentum` is a QuantConnect/LEAN repository that implements five variants (Strategy 0 to Strategy 4) of an intraday momentum strategy on SPY, backtested from May 10, 2017 to May 10, 2025 (2,012 NYSE sessions) with an initial capital of 100,000 dollars. The design is in the lineage of Zarattini, Aziz and Barbon (2024), which the paper acknowledges. Every number in this report refers to that window. The common core, identical in all five, is a breakout threshold recomputed every minute of the session: an upper band and a lower band derived from the 14-day moving average of the absolute excursions from the open, and a vol-targeted leverage

$$\Lambda = \min\left(2, \frac{0.02}{\sigma_{14g}}\right)$$

that sizes the exposure to target a daily portfolio volatility of 2%. Every position is forcibly liquidated at 15:58 ET.

The five variants differ along three axes: whether entry requires only the band break or also confirmation from a trend filter (a 100-period EMA and, in S2 only, the VWAP as well, a third condition absent from both published descriptions; see §4.1); the cadence at which the exit is checked, every 30 minutes or every 5; and whether the exit fires on the first violation or only after the violation has persisted for $N = 4$ consecutive checks. The paper presents them as a progression: S1 speeds up the exit check, S2 adds the trend filter, S3 adds persistence, and S4 combines them all. It reads the decline from S0 to S1 (return from 196% to 177%, win rate from 40% to 32%, with the Sharpe ratio almost unchanged) as what it calls the Momentum Paradox: a more aggressive risk control, without a filter on noise, destroys the return it is meant to protect. The comparison that closes the paper is instead between S4 and the two single-mechanism configurations, S2 and S3 (Table 7): S4, at 259% with a Sharpe of 1.036, is presented as the composition of the two mechanisms and described as Pareto-dominant.

This reading rests on an implicit assumption: that each row of the table isolates a single change relative to the row before it. §4 shows that this assumption does not hold. S2 and S4, the comparison on which the paper builds its final conclusion, differ on **two** axes at once, not

one; and cadence and persistence, introduced together in the step from S0/S2 to S3/S4, turn out to be separable in the available data, which is not what the design suggests at first sight.

It also rests on a point of comparison. The paper adopts only one: SPY bought and held, which over the window returns 166.4% with a Sharpe of 0.47 and a drawdown of 33.7%, and which S0 exceeds with its 196% (paper, §3.2). The comparison is between a leveraged strategy, whose leverage rule averages 1.81, and a benchmark at leverage 1. That is the point from which §2 starts.

Before going into the substance, we run an elementary check against the strongest source available: not the rounded table in the README but `stats/strat{0..4}_8y.json`, the authors' own backtest output, preserved in the repository.

T1. The five published backtests: authors' result files vs. reproduced today (tight variant, 0 bps)

	source	return	Sharpe	drawdown	win rate	orders	PSR	turnover
S0	authors	196.232%	0.835	11.2%	40%	3594	61.589%	222.35%
	reproduced	196.125%	0.835	11.2%	40%	3594	21.510%	222.36%
S1	authors	177.757%	0.842	10.1%	32%	4900	67.736%	303.52%
	reproduced	177.468%	0.841	10.1%	32%	4900	21.982%	303.52%
S2	authors	227.320%	0.948	8.7%	40%	3450	75.319%	213.01%
	reproduced	227.196%	0.947	8.7%	40%	3450	33.461%	213.01%
S3	authors	252.248%	1.012	9.3%	40%	3550	81.164%	219.91%
	reproduced	252.101%	1.011	9.3%	40%	3550	41.219%	219.91%
S4	authors	259.235%	1.036	8.4%	40%	3388	83.439%	209.40%
	reproduced	259.177%	1.036	8.4%	40%	3388	44.342%	209.40%

Authors: `stats/strat{s}_8y.json`, the backtest output preserved in the repository. Reproduced: the same code re-run today. Orders, Sharpe, drawdown, win rate and turnover agree within rounding on all five rows; the PSR differs by 39.1 to 45.8 points (41.3 on average), because the platform now subtracts the risk-free rate inside it and in March it did not. See §1 and §7.

Orders, drawdown and win rate match exactly on all five rows; Sharpe matches to within a thousandth, turnover to within a hundredth of a point. Return matches to within 0.29 percentage points (pp) in the worst case (S1) and 0.06pp in the best (S4), always on the low side. The gap has an identifiable cause: at exactly the same order count, the commissions of the new runs are higher by between 0.24% and 0.51%, so the difference lies in the fill prices, not in the strategy. The likely source is the dividend adjustment. LEAN fills on dividend-adjusted prices, and the adjustment is computed as of the day the backtest runs, so every

dividend paid between two runs lowers the whole price history seen by the later one, by about a quarter of a percent for SPY. Lower prices buy more shares for the same notional, and the per-share commission, with the drag it puts on return, rises by the same fraction. SPY went ex-dividend twice between the authors' runs and these, on 20 March and 18 June 2026. The runs of 18 and 20 March (S0 to S3), consistent with data that did not yet reflect the first of the two, differ from the reproduction by 0.47% to 0.51%, two dividends; S4, run on 23 March, differs by 0.24%, one. This is inferred from the dates and the magnitudes, not verified against the platform's price files. The code in the repository does today what it did for the authors.

One exception, and only one: the Probabilistic Sharpe Ratio does not reproduce on any of the five rows, with a gap of between 39.1 and 45.8 percentage points, 41.3 on average. This is not a code reproducibility problem: the platform has changed how it computes the statistic. LEAN's own routine returns the authors' values when the Sharpe ratio inside it is taken before the risk-free rate, and today's when the rate is subtracted, to within 0.3 points on all ten figures (§7). The discussion is in §7 and in Appendix A, where it becomes the reason for computing the Deflated Sharpe Ratio from the raw moments rather than citing it.

Beyond that, the issue this report takes up is not the reproducibility of the numbers. It is what those numbers allow one to conclude, and that is the question the sections that follow answer.

1.1 Conventions and units

Three different quantities appear in what follows, and they are fixed here, because confusing them changes the conclusions by a factor of three.

Return. Unless otherwise stated, "return" is the cumulative return over the full eight-year window, net of commissions, as QuantConnect reports it: the 196.125% of S0 means that 100,000 dollars become 296,125.

Log return. When *effects* are compared (how much an entry condition is worth, how much a cadence costs), the metric is $\ln(1 + R)$, and differences between cells are differences of logarithms. A value of -0.06508 means that final wealth is $e^{-0.06508} = 93.7\%$ of the reference's, that is, 6.3% less; on cumulative return the same gap is worth 18.7 percentage points, because the return is compounded over eight years. The two readings must not be mixed.

The choice of the logarithm is not cosmetic. The cells of the design start from different levels, 196% under `tight` and 229% under `loose`, and only a proportional measure makes them

comparable. Above all, the factorial design of §4 rests on tests of the form "the interaction is zero, therefore the two effects are additive": if two independent effects multiply wealth by $(1 + a)$ and $(1 + b)$, in percentage points the combined effect is $a + b + ab$, and a spurious interaction equal to the product would appear; with the values at stake, about ten times the one actually measured. In logarithms, independent effects add by construction, so an interaction that is zero to the fourth decimal place means something.

Basis points (bps) of terminal log wealth. In §3 the residuals of the cost model are measured in this unit: 1 bps is 0.01% of the final multiplier. It is needed because the same error in percentage points weighs differently on a run that closes at +250% and on one that closes at -53%, so in percentage points the residuals would not be comparable across rows.

Costs. The horizontal axis of every cost chart and cost table is the **additional** slippage: the 0 bps level is not a backtest without costs, it is a backtest that pays Interactive Brokers commissions in full and the quoted bid-ask spread, since LEAN fills buys at the ask and sells at the bid, and nothing else. The risk-free rate is set at $r_f = 2\%$ and the margin financing rate at 3.5%, both conventions declared in advance and discussed where they are used.

§2. What the strategy is measured against

A return of 271% over eight years means nothing until one says what it is measured against. The paper does adopt a point of comparison, SPY bought and held at 166.4%, but without leverage, while the strategy runs at an average leverage of 1.81: the comparison is between a leveraged portfolio and one that is not. The Sharpe ratio attached to that benchmark, 0.47, is on the same footing as the strategy's: under LEAN's convention (Appendix A, §A.8), reading the 16% volatility stated on the same line as the platform's annualized figure, 166.4% over eight years gives exactly 0.47 at the platform's risk-free rate of about 2.7%. The "nearly double" that the paper states for S0, 0.835 against 0.47, is therefore a like-for-like comparison. The textbook formula, $(CAR - r_f)/\sigma$ at the report's $r_f = 2\%$, would give 0.69 for the benchmark, but also about 1.4 for S0: applied consistently, either convention returns a ratio between 1.8 and 2.0. What the comparison does not match is leverage. This section builds two benchmarks in its place, each designed to isolate one thing, before going into the construction itself. They are what determines which questions are worth asking afterwards.

Both benchmarks share the strategy's leverage rule, $\Lambda = \min(2, 0.02/\sigma_{14g})$, and differ only in what they are meant to isolate. Leverage is matched because it is a free parameter that anyone can raise: comparing a portfolio at leverage 1.81 with one at leverage 1 measures the signal and the decision to take more risk together, and does not separate the two. And the

whole rule is matched, not just the average multiplier, because the rule has effects of its own on compounded return. Volatility targeting cuts leverage on the most volatile days, where leverage costs most, so it should pay less drag than a fixed leverage of the same mean; whatever that advantage is worth, it belongs to the null hypothesis, not to the signal.

One thing about that rule changes how it needs to be understood. On this window and with the repository's parameters, Λ is **at the cap of 2 on 66.1% of days**, with an average of 1.8056: for two thirds of the time there is no targeting at all, only fixed leverage at the maximum allowed. The figure comes from the instrumented benchmark, which replicates the rule without a signal, but it does not depend on the benchmark: the average weight per fill of the sixteen strategy configurations, $\bar{w} \in [1.805; 1.828]$ in §3.7, is the same average leverage read off their own orders. The cap is not incidental: the paper sets it at 2× for consistency with Zarattini et al. and notes (§2.3) that without it, leverage in quiet periods exceeded the available margin and triggered insufficient buying power errors.

The first benchmark, *intraday*, is long SPY from 9:31 to 15:58 and closes every evening, with no signal at all: it is the strategy stripped of nothing but its entry and exit logic. The second, *hold24*, keeps the position overnight as well: it is leveraged buy-and-hold, that is, the alternative an investor actually has available. Both are needed because they answer two different questions and neither is sufficient on its own. *hold24* says whether it would have been better to do something else, but it differs from the strategy on three axes at once (trading window, direction selection, entry timing) and therefore cannot attribute anything to the signal. *intraday* differs on one axis only, the presence of the signal, and is therefore the only one of the two that isolates it.

T2. The two full-session benchmarks (entry at 9:31) vs. the best configuration

	intraday (9:31 to 15:58)	hold24	S4' tight
net profit	-18.31%	350.52%	271.37%
CAR	-2.50%	20.70%	17.82%
Sharpe (QC)	-0.154	0.615	1.075
drawdown	40.5%	40.2%	9.0%
recovery (days)	147	714	343
β	0.800	1.323	-0.053
α	-0.089	0.043	0.101
PSR	0.004%	6.915%	49.490%
orders	3,997	1,587	3,336

	intraday (9:31 to 15:58)	hold24	S4' tight
commissions (USD)	9,746.60	1,604.39	15,708.03
w_{sum}	7,171.26	68.89	6,024.57
b^* (bps)	-0.282	218.486	2.178

There is a signal

The full-session benchmark, the one entering at 9:31 and reported in T2, loses 18.31%, with a Sharpe of -0.154 and a drawdown of 40.5%. Matched to the strategy's entry time it loses far more, but the point already holds in this form.

This is the most important result of the section: before this run, none of the 54 backtests collected up to that point excluded the possibility that the strategy's return came simply from being long intraday at leverage 1.8.

Instrument and leverage rule are identical; the trading window is not, and the difference was measured rather than argued. The benchmark enters at 9:31, while the strategy cannot enter before 10:00: the thirty-minute cadence places the first usable evaluation there, as the paper itself notes in a footnote. Repeating the run with entry at 10:00 and nothing else changed, the matched benchmark closes at -36.08% , not at -18.31% . That it is the same experiment with one variable moved is confirmed by the controls: $\lambda_{avg} = 1.8056$ and $pct_capped = 66.1\%$ identical to the last digit, w_{sum} within 0.02%, turnover within 0.01%, order count within four out of four thousand.

The timing difference was therefore worth 17.8 points, and in the opposite direction to the one expected on paper: the half-hour between 9:31 and 10:00 works in favor of whoever holds it, and letting the benchmark keep it was **helping** the benchmark, not penalizing it. The correctly matched comparison, same window as the strategy, same leverage, no signal, puts $+271.37\%$ vs. the -36.08% of the long null and the -30.75% of the short null (next paragraph): **between 302 and 307 points depending on the direction in which the null hypothesis is held**, vs. the 289.7 of the unmatched comparison. The number to cite is the narrower of the two, 302.1, because the strategy trades in both directions and the null should be taken in its most generous version. This is also a first observation about the design of the repository, not only a methodological qualification: the thirty-minute entry cadence excludes the opening half-hour by construction, and in this sample that half-hour contributes positively while the rest of the session does not, and it is worth 17.8 points to a passive leveraged position.

T3. The benchmark matched on entry time (10:00, as the strategy)

Same script, same leverage rule, one variable changed: the entry minute, and for the third column the sign of the position.

	entry 9:31, long	entry 10:00, long	entry 10:00, short
net profit	-18.31%	-36.08%	-30.75%
Sharpe (QC)	-0.154	-0.299	-0.260
drawdown	40.5%	49.0%	46.4%
β	0.800	0.724	-0.724
α	-0.089	-0.106	0.011
avg win / avg loss	0.93% / -1.06%	0.83% / -0.99%	0.99% / -0.83%
orders	3,997	4,001	4,004
commissions (USD)	9,746.60	8,869.29	9,799.29
ann_std	0.182	0.170	0.170
w_{sum}	7,171.26	7,170.04	7,171.86
turnover	245.45%	245.42%	245.45%
lambda_avg	1.8056	1.8056	1.8056
pct_capped	66.1%	66.1%	66.1%

Controls: `lambda_avg` and `pct_capped` identical to the last digit on all three runs (same leverage path), w_{sum} within 0.03%, $w_{sum}/(\text{turn}/100) \in [2,921.5; 2,921.9]$ vs. the 2,922 calendar days. The long and the short at 10:00 are mirror images in β (± 0.724) and have average win and average loss swapped: they are the same exposure in the two directions.

A natural objection is that the benchmark is long-only while the strategy also goes short, so the comparison is not matched on exposure. This too was measured rather than estimated: the same benchmark, same 10:00 to 15:58 window, with only the sign of the position reversed, closes at -30.75% . It is the exact mirror of the long run, with $\beta = -0.724$ vs. $+0.724$ and average win and average loss swapped digit for digit ($0.99\% / -0.83\%$ vs. $0.83\% / -0.99\%$), so the two rows measure the same exposure in the two directions.

Blind exposure loses in both directions, and the gap between the two is 5.3 points over eight years. That gap, and only that, comes from the drift of the window, which enters the two runs with opposite signs, so the drift itself is worth about half of it; the rest, more than thirty points and common to both, is variance drag, commissions and the bid-ask spread, which depend on Λ^2 and on the number of fills and do not distinguish the sign of the position. The asymmetry exists and favors the short side, not the long side as one would have expected, but it is an order of magnitude too small to make either direction a sensible choice. There is no right side to be on without a signal: it is the cost of *being there*, not the direction, that determines the outcome.

On the opposite side, the trade files published in the repository rule out the possibility that the edge lives on the short side: on S4 the 878 long trades produce 137,578 dollars and the 816 short trades 137,500, that is, 50.0% and 50.0% of the P&L, and the split is close to even on S2 and S3 as well (49.6% and 48.8% long). The P&L is in dollars not rescaled by the equity of the moment, but the two directions are interleaved in time, so the imbalance favors neither. It is the same fact that can be read in $\beta = -0.053$: if the strategy were predominantly long, its beta would be a positive fraction of the intraday benchmark's 0.800.

Why the intraday benchmark loses

Not because SPY falls: over the same window `hold24` makes +350.5%. It loses because of two mechanical effects, both independent of the direction of the market. The decomposition that follows is that of the full-session run, entering at 9:31, the only one of the three instrumented with the accumulators, and its items should be understood relative to that –18.31%, not to the matched –36.08%: the mechanism is the same, the magnitudes are not.

Both effects are measured, not estimated. The obvious route, deriving them from the statistics LEAN reports, would make them inherit LEAN's conventions, and Appendix A, §A.8 shows that at least one of those conventions is not the one a reader would assume. They are therefore collected inside the backtest: a single run at 0 bps, with four scalar accumulators updated at every session close and emitted via `set_runtime_statistic`, the only channel on the Free plan that does not consume the log quota. The number of samples comes back as 2,012, exactly the independent count of NYSE sessions in the window, a free check that the sampling is one point per session with no gaps.

Commissions: 9,746.59 dollars on an account that starts at 100,000 and closes at 81,691, with 3,997 orders over eight years. The right denominator is not the initial capital. The Interactive Brokers fee model is per share, and a trade buys $\Delta E/P$ shares: the commission is $c \Delta E/P$, and relative to equity it is $f = c \Delta/P$, a quantity that does not depend on the size of the account. Summed over the sessions, the dollars paid are $\sum_t f_t E_t \approx f \sum_t E_t$: they are proportional to the **integral of equity over time**, not to the starting capital, and the annual rate is total commissions divided by that integral. Dividing by 100,000 instead would understate it, because the account shrank over the whole window. The run records the equity at every session close, 2,012 values with an average of 87,027, so the integral is a sum with one step per session, $\int_0^8 E_t dt \approx 8 \bar{E} = 696,215$, and

$$\phi = \frac{\text{commissions}}{\text{years} \cdot \bar{E}} = \frac{9,746.59}{8 \cdot 87,027} = 1.400\% \text{ per year}$$

The step $f_t \approx f$ treats the rate as constant; strictly, the ratio gives its average weighted by equity.

Leverage drag. Holding Λ times the exposure does not multiply the compounded return by Λ . Over a single session the log return of the leveraged position is $\ln(1 + \Lambda r) \approx \Lambda r - \frac{1}{2}\Lambda^2 r^2$, while Λ times the log return of the underlying is $\Lambda r - \frac{1}{2}\Lambda r^2$: the difference, what leverage costs **beyond** simple rescaling, is $\frac{1}{2}\Lambda(\Lambda - 1)r^2$. Since the portfolio return is $r_{port} = \Lambda r$, the same quantity can be written with the portfolio series alone:

$$\frac{1}{2}\Lambda_t(\Lambda_t - 1)r_t^2 = \frac{1}{2}\frac{\Lambda_t - 1}{\Lambda_t}r_{port,t}^2$$

and this is the form accumulated session by session inside the run. The sum built this way never passes through `ann_std`, so it does not inherit LEAN's annualization convention (Appendix A, §A.8). Divided by the number of years, it is 0.969% per year. The accumulator adds the term only on sessions with $\Lambda_t > 1$: the guard is there to skip sessions without a position, where $\Lambda_t = 0$, but it also drops the few with $0 < \Lambda_t < 1$, when the trailing volatility exceeds 2% a day. On those the term is negative, since a leverage below one recovers part of the variance instead of paying it, so the 0.969% slightly overstates the drag.

The arithmetic. The two drags are logarithmic and add to each other, so they must be compared with the log CAR, $-\ln(1 + R)/8 = 2.528\%$, and not with the simple CAR; mixing the two scales is worth a few tenths of a point:

$$1.400\% + 0.969\% = 2.369\% \quad \text{vs.} \quad 2.528\%$$

T4. Decomposition of the `intraday` benchmark's return (entry at 9:31, 0 bps)

item	how it is obtained	value
commissions	$\phi = \text{fees}/(\text{years} \cdot \bar{E})$	1.400%/year
leverage drag	$\frac{1}{2} \sum_t \frac{\Lambda_t - 1}{\Lambda_t} r_{port,t}^2 / \text{years}$	0.969%/year
sum of the two mechanical effects		2.369%/year
measured log CAR	$-\ln(1 + R)/\text{years}$	2.528%/year
residual, leveraged position		-0.159%/year
residual, underlying	residual/ $\bar{\Lambda}$	-0.088%/year

Raw values from the instrumented run (Runtime Statistics, not recomputable from a CSV, since the Free plan offers no API access): `eq_n` = 2012, `avg_equity` = 87026.93, `drag_sum` = 1.550661e-01, `total_fees` = 9746.59. The `eq_n` count coincides with the NYSE sessions counted separately in the window: an independent check that the sampling is one point per session, with no gaps.

The two mechanical effects therefore explain 94% of the loss. What remains, -0.159% per year, is the compounded return of the position itself once commissions and the extra cost of leverage are removed; divided by the average leverage it is about -0.09% per year for an unleveraged position. The division is only approximate, because leverage changes from day to day and is not independent of the returns it multiplies, so the figure is an order of magnitude and nothing below rests on its second digit.

In words: in this window, anyone long SPY from 9:31 to 15:58 every day, without leverage and commissions but paying the bid-ask spread, would have ended roughly flat after eight years. Trading hours, on their own, earned about nothing: the 13.0% a year that SPY delivered (166.4%, paper, \$3.2) accrued almost entirely overnight, the segment this benchmark does not hold, in line with a well-documented pattern in US equities (Cooper, Cliff and Gulen, 2008; Lou, Polk and Skouras, 2019). This is why the benchmark loses: with nothing to earn during the session, what it pays in commissions and leverage drag is a net loss, and whatever the strategy earns over it has to come from its signal.

"Flat" refers to the compounded return: the arithmetic mean of the daily returns is positive, but compounding subtracts about half their variance.

Averaging $\ln(1 + r) \approx r - \frac{1}{2}r^2$ gives $E[\ln(1 + r)] \approx E[r] - \frac{1}{2}E[r^2] = E[r] - \frac{1}{2}(\sigma^2 + E[r]^2) \approx E[r] - \frac{1}{2}\sigma^2$, since $E[r]^2$ is several orders of magnitude smaller than σ^2 for daily returns.

From this follows the property that makes the strategy interesting. A strategy that goes flat every evening does not touch the equity premium by construction: it extracts return from a window whose compounded return is zero. Its beta of -0.053 has a different origin, and the benchmark shows which: the same schedule held long, with no signal, has $\beta = 0.800$. Being flat overnight removes the equity premium, not the exposure to the day's move. What removes that is the signal, which is long on upward breakouts and short on downward ones and splits its P&L evenly between the two sides (50.0% and 50.0% on S4, above), so that its daily return depends on the size of the intraday move more than on its sign. The zero beta is not the result of an optimization; it is a consequence of the symmetry of the signal. The schedule, for its part, has five exceptions in eight years, all on half-day sessions where the 15:58 liquidation branch is never reached (\$7, item 3); \$8 bounds what they are worth.

The opportunity cost

The uncomfortable question remains. `hold24` makes $+350.5\%$ vs. $+271.4\%$, and it is almost immune to costs: its break-even is at 218 bps of slippage vs. the strategy's 2.2, because `set_holdings` rebalances only the delta and the notional traded is therefore two orders of magnitude smaller. Under any cost assumption the gap widens rather than closes.

One correction, however, works in the opposite direction. LEAN does not charge interest on the borrowed balance, and with an average leverage $\bar{\Lambda} = 1.8056$ hold24 borrows 0.8056 units of equity every night for eight years. A broker would charge interest on that balance, and the interest depends on how much is borrowed, not on how the market moves. Write E_t for the equity at the start of year t , $g = 20.702\%$ for the annual return of hold24 as backtested (its CAR, before any financing cost) and r for the annual financing rate. The borrowed balance is $(\bar{\Lambda} - 1)E_t$, the interest paid in the year is $(\bar{\Lambda} - 1)r E_t$, and

$$E_{t+1} = E_t (1 + g - (\bar{\Lambda} - 1)r), \quad E_8 = E_0 (1 + g - (\bar{\Lambda} - 1)r)^8.$$

The cost is therefore subtracted from the annual growth rate, not applied as a fraction of the return. Using the average leverage and a constant rate are both simplifications; T5 applies the correction for rates from 2% to 5%.

T5. Correction for the financing that LEAN does not charge (hold24, gross CAR 20.702%)

r	annual cost	hold24 corrected
2.0%	1.61%	304.6%
2.5%	2.01%	293.8%
3.0%	2.42%	283.2%
3.5%	2.82%	272.9%
4.0%	3.22%	262.8%
4.5%	3.63%	253.0%
5.0%	4.03%	243.4%

Subtractive form, borrowed balance $(\Lambda - 1) = 0.8056$. At the chosen rate (3.5%) hold24 makes 272.9% vs. 271.4% for S4' tight. Break-even rate: 3.58%.

At the declared convention, $r = 3.5\%$ (average T-bill rate over the window plus a retail-account spread), hold24 makes 272.9% vs. 271.4%. The rate at which the two break even is 3.58%, that is, inside the range the sensitivity table covers.

The asymmetry deserves stating, otherwise the correction looks chosen for convenience: it hits hold24 in full and barely touches the strategy, because brokers charge interest on the overnight balance and the strategy is flat every evening.

But the conclusion that holds is not that the strategy wins. It is that the return gap between the two is of the same order as the uncertainty on the one cost item the backtest does not charge, and therefore does not support an argument in either direction.

What remains

The comparison that holds is the other one, and it does not depend on r . Same instrument, same window, same leverage rule: drawdown 9.0% vs. 40.2%, $\beta = -0.053$ vs. 1.323, $\alpha = +0.101$ vs. $+0.043$, PSR 49.5% vs. 6.9%. Recovery from the maximum drawdown takes 343 days vs. 714. The two PSRs are measured in the same session and on the same version of the platform, so they are comparable with each other; §7 explains why they should not be compared with the published ones.

The defensible formulation is therefore this: it is not a better way of earning from the market, it is a source of return almost orthogonal to the market at a quarter of the drawdown. "271 vs. 350" is the less informative comparison, and the columns that show it are above.

The rest of the report answers a different question, one that only makes sense after this one: given that this is a zero-beta diversifier, is it well built?

§3. Method: one backtest, every cost level

With this code architecture, a single instrumented backtest at 0 bps contains every cost level in closed form.

The question this section answers, how much execution cost the strategy can absorb before its return is gone, is not invented for the occasion: the paper raises it itself. Among the lines of future work listed in §6.7 is "re-running with 2× and 5× baseline transaction costs to establish the break-even cost level". What follows answers it for every cost level rather than for two, from the single 0 bps run; the forty runs at higher costs in §3.6 serve only to check the answer.

3.1 The two invariances everything depends on

In an arbitrary backtest, introducing costs changes the history: the capital available at each instant is different, and if decisions depend on capital or on execution prices, then *which* trades occur changes as well. When that happens there is no closed form: every cost level is a different experiment and has to be re-run.

The repository's code has two properties that break that circle.

The first: **the signal does not look at execution prices**. Entries and exits in all five strategies compare `data[symbol].close`, the market price of the bar, with bands and averages

computed from market prices. No rule refers to the price at which an earlier order was filled, as a stop-loss measured from the entry price would. Slippage changes fill prices and leaves market prices intact, so it cannot change when a position opens or closes.

The second: **sizing is relative to equity**. `set_holdings` receives a fraction of net equity, Λ_t , not a number of shares, and buys $\Lambda_t E/P$ shares. A smaller account buys proportionally fewer shares, but the position remains the same fraction of capital. With a fixed number of shares instead, an account reduced by costs would hold a larger fraction of its capital, and the costs would change the exposure itself.

It follows that the set of trade instants $T = \{t_1, \dots, t_N\}$, the direction of each, and the weight of each as a fraction of equity are **identical at every cost level**. What changes is how much is subtracted at each fill, not which fills occur. This is a property of the code, not of the data, and it is falsifiable: a stop-loss anchored to the fill price, or sizing by a fixed number of shares, would be enough to break it. The constant order count across the five cost levels, in all ten combinations, is the check that it does not break here.

3.2 The equity curve at cost b

Let $E_0(t)$ be the equity path in the 0 bps run. For the i -th fill, executed at t_i , let Q_i be the number of shares, P_i the market price and V_i the equity immediately before the order. The **execution weight** is the notional traded as a fraction of equity:

$$w_i = \frac{|Q_i P_i|}{V_i}$$

With an additional slippage of b basis points ($1\text{bps} = 10^{-4}$), the fill loses a fraction $w_i b/10^4$ of equity. Since the effect is multiplicative and compounds fill after fill:

$$E_b(t) = E_0(t) \cdot \prod_{i: t_i \leq t} \left(1 - \frac{w_i b}{10^4}\right)$$

Under the two invariances of §3.1 this is not an approximation: it is an identity, because the product runs over exactly the same fills with exactly the same weights.

3.3 First order and break-even

Moving to the cumulative return $1 + R_b = E_b(T)/E(0)$ and taking the logarithm, the product becomes a sum:

$$\ln(1 + R_b) = \ln(1 + R_0) + \sum_{i=1}^N \ln\left(1 - \frac{w_i b}{10^4}\right)$$

The cost per single fill is of order 10^{-4} , so the expansion $\ln(1 - x) = -x + O(x^2)$ is extremely accurate term by term:

$$\ln(1 + R_b) = \ln(1 + R_0) - \frac{b}{10^4} w_{sum} + O(b^2), \quad w_{sum} = \sum_{i=1}^N w_i$$

The log return is **affine in b** , with slope $-w_{sum}/10^4$. A single quantity accumulated during the run, w_{sum} , describes the entire cost map. Setting the return to zero gives the break-even level:

$$b^* = \frac{10^4 \ln(1 + R_0)}{w_{sum}}$$

The neglected term is $-\frac{b^2}{2 \cdot 10^8} \sum_i w_i^2$. The sum of squares is not observable from the data collected; w_{sum}^2/N is a lower bound for it by Cauchy-Schwarz, with equality if all weights were identical. It is used further on to verify that the residual grows where it should.

3.4 Where the commissions are

The model reconstructs **slippage**. Commissions are not modeled at all: they are already inside E_0 , because the 0 bps run is not a run without costs but a run without *additional* slippage, one that pays the Interactive Brokers fee model in full and the quoted bid-ask spread (\$1.1). The horizontal axis of every chart is therefore "additional slippage", not "total cost".

For this to work only one thing is needed: that the commission, as a **fraction of equity**, be the same at every level of b for the same trade. The fee model is per share, with a rate of $c = 0.005$ per share, and the quantity of an order is $|Q_i| = \Lambda_i V_i / P_i$. Therefore

$$\frac{\text{Fee}_i}{V_i} = c \frac{\Lambda_i}{P_i}$$

and both Λ_i (which comes from SPY's 14-day volatility, a market quantity) and P_i are **invariant in b** . The commission takes the same fraction of equity at every cost level: it multiplies every path by the same sequence of factors, and cancels in the ratio E_b/E_0 . This does not require the fraction to be constant *over time*, and it is not: it goes as $1/P_i$, so over the sample it falls to about 37% of its initial value while the price SPY trades at in the backtest, adjusted for dividends, moves from about 207 to about 564 dollars. What the model needs is that the fraction be the same in the two runs **at the same instant**, and it is, because at that instant both see the same price and the same leverage. On the order of magnitude, half a cent per share is equivalent to a tenth to a quarter of a basis point of slippage, a fraction of what is being measured on the cost axis.

That commissions do follow equity can be seen in the raw data. Between the S0 `tight` run at 0 bps and the one at 1 bps, commissions paid go from 15,460 to 11,230 dollars, that is, to 73%, at exactly the same 3,594 orders; final equity instead falls to 52%. They are neither fixed nor proportional to the final result: they follow equity along the whole path, higher at the start and lower at the end. The real check, however, is the out-of-sample validation of §3.6, whose forty predicted returns are all net of commissions: if commissions did not follow equity, the smaller account at high b would pay a larger fraction of them, and the model would overstate returns by an amount increasing in b . It does not.

3.5 The per-order minimum

The declared fee model also has a minimum of 1 dollar per order, and a fixed minimum is not proportional to equity. That is not the regime in operation: the average order pays between 3.98 and 4.77 dollars across all sixteen configurations, four to five times the threshold, and not even the most penalized run gets there, since S1 `tight` at 2 bps closes at -53% and still pays 1.87 dollars per order. Where the minimum could bite, on the most de-leveraged days, it would make commissions slightly *more* than proportional: the error is bounded and of known sign. The one run that sits entirely inside the minimum is `hold24`, at 1.011 dollars per order, because it rebalances only the delta and its orders are two orders of magnitude smaller: one more reason, independent of those in §2, why its cost structure is not comparable with that of the strategies.

3.6 Out-of-sample validation

Forty rows, the four at $b > 0$ for each of the ten strategy/threshold combinations, none of them used to calibrate the model, which starts only from the 0 bps run of each combination.

T6. Out-of-sample validation of the cost reconstruction

	exit threshold	bps	measured	predicted	error (pp)
S0	<code>tight</code>	0.25	151.688	151.706	+0.018
S0	<code>tight</code>	0.50	113.854	113.949	+0.095
S0	<code>tight</code>	1.00	54.578	54.578	-0.000
S0	<code>tight</code>	2.00	-19.349	-19.310	+0.039
S0	<code>loose</code>	0.25	185.827	185.793	-0.034
S0	<code>loose</code>	0.50	148.064	148.115	+0.051
S0	<code>loose</code>	1.00	87.004	87.007	+0.003

	exit threshold	bps	measured	predicted	error (pp)
S0	loose	2.00	6.195	6.235	+0.040
S1	tight	0.25	122.325	122.263	-0.062
S1	tight	0.50	78.042	78.041	-0.001
S1	tight	1.00	14.231	14.242	+0.011
S1	tight	2.00	-53.148	-52.963	+0.185
S1	loose	0.25	162.825	162.831	+0.006
S1	loose	0.50	120.309	120.283	-0.026
S1	loose	1.00	54.707	54.734	+0.027
S1	loose	2.00	-23.660	-23.652	+0.008
S2	tight	0.25	180.068	180.019	-0.049
S2	tight	0.50	139.657	139.644	-0.013
S2	tight	1.00	75.472	75.519	+0.047
S2	tight	2.00	-5.885	-5.845	+0.040
S2	loose	0.25	214.209	214.304	+0.095
S2	loose	0.50	174.561	174.629	+0.068
S2	loose	1.00	109.633	109.672	+0.039
S2	loose	2.00	22.150	22.216	+0.066
S3	tight	0.25	199.832	199.812	-0.020
S3	tight	0.50	155.270	155.288	+0.018
S3	tight	1.00	85.043	85.095	+0.052
S3	tight	2.00	-2.754	-2.698	+0.056
S3	loose	0.25	200.875	200.867	-0.008
S3	loose	0.50	161.786	161.781	-0.005
S3	loose	1.00	98.235	98.183	-0.052
S3	loose	2.00	13.540	13.585	+0.045
S4	tight	0.25	208.207	208.197	-0.010
S4	tight	0.50	164.435	164.454	+0.019

	exit threshold	bps	measured	predicted	error (pp)
S4	tight	1.00	94.664	94.711	+0.047
S4	tight	2.00	5.491	5.553	+0.062
S4	loose	0.25	205.448	205.438	-0.010
S4	loose	0.50	167.719	167.647	-0.072
S4	loose	1.00	105.516	105.514	-0.002
S4	loose	2.00	21.159	21.171	+0.012

Eight-year net profit (%). 40 rows, none used to calibrate the model: calibration uses only the 0 bps run of each combination. The reading of the residuals is in §3.6.

$\sum_i w_i^2$ is not observable from the data collected; the expected term uses w_{sum}^2/orders , which by Cauchy-Schwarz is a lower bound for it, with equality if leverage never varied. The comparison is therefore one of order of magnitude, not a test.

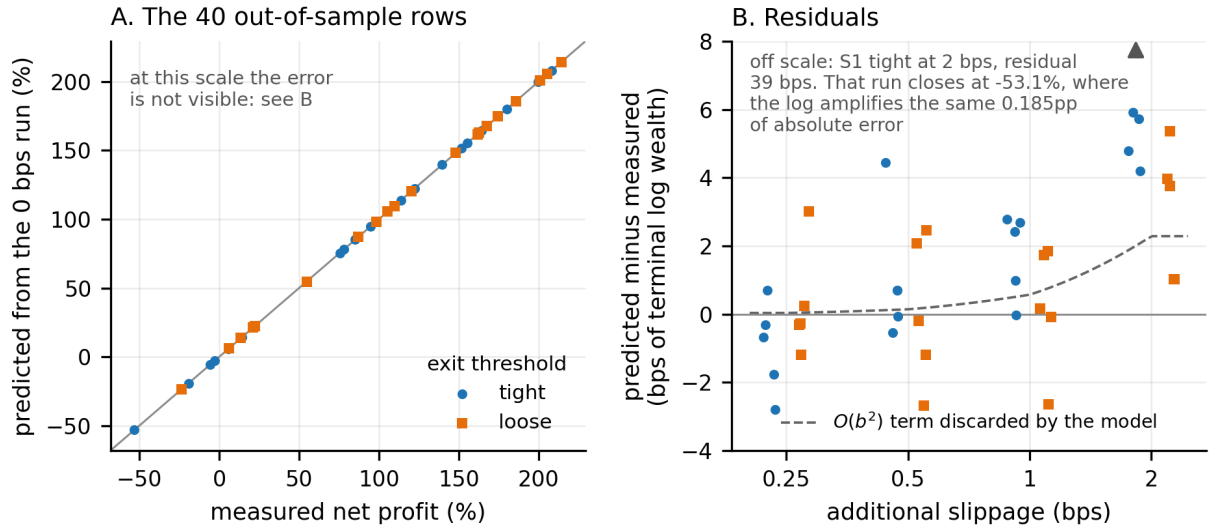
Maximum error 0.185 percentage points (pp), **mean absolute error 0.038pp**. The order count stays constant across the five levels everywhere, so the invariance of §3.1 is not assumed but verified.

In percentage points, however, the error depends on the scale of the return and is not comparable across rows: the invariant measure is the residual in bps of terminal log wealth. Up to 1 bps the mean residual is +0.41 bps with a dispersion of 1.79 and no pattern, that is, noise. At 2 bps the mean rises to +3.98 bps over the nine regular rows, compared with the $O(b^2)$ term of about 2.19 bps expected from the lower bound of §3.3: same sign, same order of magnitude. The linear model begins to understate the cost exactly where it should.

The one off-scale row, 39 bps, is S1 tight at 2 bps, a run that closes at −53.1%: near the point where capital goes to zero the logarithm amplifies, and the same 0.185pp in absolute terms weighs far more. This is not a failure of the model; it is the model's natural measure blowing up where wealth goes to zero.

3.7 The $\bar{w} \approx \Lambda$ relation

The average weight per order, $\bar{w} = w_{sum}/N$, lies between 1.8048 and 1.8282 across all sixteen configurations, vs. an average leverage `lambda_avg` = 1.8056.



F1. Out-of-sample validation of the cost reconstruction

T7. The $\bar{w} \approx \Lambda$ relation, break-even and ratio to turnover

	variant	orders	w_{sum}	$\bar{w}$	b^* (bps)	$w_{sum}/(\text{turn}/100)$
S0	tight	3,594	6,500.8574	1.8088	1.670	2,923.6
S0	loose	3,116	5,654.9449	1.8148	2.107	2,923.7
S1	tight	4,900	8,873.8389	1.8110	1.150	2,923.6
S1	loose	3,864	7,064.0801	1.8282	1.618	2,923.8
S2	tight	3,450	6,228.1092	1.8052	1.903	2,923.9
S2	loose	2,982	5,397.5556	1.8100	2.372	2,923.9
S3	tight	3,550	6,430.5101	1.8114	1.957	2,924.2
S3	loose	3,060	5,566.3676	1.8191	2.229	2,924.0
S4	tight	3,388	6,122.9884	1.8073	2.088	2,924.1
S4	loose	2,914	5,283.0896	1.8130	2.363	2,923.8
S2'	tight	3,498	6,318.6135	1.8064	1.822	2,923.8
S2'	loose	3,026	5,480.9952	1.8113	2.284	2,923.8
S3'	tight	3,374	6,089.4032	1.8048	2.080	2,924.1
S3'	loose	2,886	5,228.7979	1.8118	2.385	2,923.9
S4'	tight	3,336	6,024.5691	1.8059	2.178	2,924.1
S4'	loose	2,864	5,188.1921	1.8115	2.462	2,923.9

	variant	orders	w_{sum}	$\bar{w}$	b^* (bps)	$w_{sum}/(\text{turn}/100)$
bench intraday	none	3,997	7,171.2571	1.7942	-0.282	2,921.7
bench hold24	none	1,587	68.8944	0.0434	218.486	2,919.3
bench 10:00 long	none	4,001	7,170.0368	1.7921	-0.624	2,921.5
bench 10:00 short	none	4,004	7,171.8552	1.7912	-0.512	2,921.9

Over the sixteen strategy configurations $\bar{w} \in [1.8048; 1.8282]$, vs. $\text{lambda_avg} = 1.8056$. The two deviations (hold24 and the intraday benchmark) are discussed below.

Ratio to LEAN's turnover: $w_{sum}/(\text{turnover}/100) \in [2, 919.3; 2, 924.2]$ on all twenty rows, vs. 2,922 calendar days (2,012 NYSE sessions) in the window. LEAN's Portfolio Turnover is the average, over its daily samples, of the notional traded that day divided by equity, and w_{sum} is the same sum taken fill by fill; their ratio is therefore the number of samples LEAN used. On all twenty rows it is the number of calendar days, not of trading sessions: LEAN keeps one point per calendar day, with zero on days without trading, and it computes volatility, Sharpe ratio and PSR on that same series. Appendix A, §A.8 reaches the same conclusion from the moments of the returns; this check uses only the volume traded.

This is not an empirical regularity: it follows from how the strategies trade. They are exclusively intraday, so every position opens from zero and closes in full within the day. An entry moves exactly the target position, so its weight is the leverage of that day, Λ_t ; an exit closes the same shares at a different price, so its weight differs from Λ_t only by the trade's own return, $w_{exit} = \Lambda_t(1+x)/(1+\Lambda_t x)$ for a long whose price has moved by x . The mean of the weights is therefore the leverage averaged over orders. It comes close to the average over days, 1.8056, without having to equal it: a strategy that trades more often on quiet days, when Λ_t sits at the cap of 2, has a slightly higher $\bar{w}$.

Practical consequence: substituting $w_{sum} \approx 1.81 N$ into the break-even formula,

$$b^* \approx \frac{10^4 \ln(1 + R_0)}{1.81 N}$$

the cost tolerance of any strategy in this class can be computed from the README table alone, return and number of orders, before running any backtest. Checked on five configurations against the exact value from w_{sum} : S2 1.90 vs. 1.903; S3 1.96 vs. 1.957; S4 2.09 vs. 2.088; S0 tight 1.669 vs. 1.670; S0 loose 2.113 vs. 2.107.

The relation has two deviations, both expected. hold24 has $\bar{w} = 0.0434$, because `set_holdings` in that case rebalances only the delta relative to the position already open: it neither opens nor closes whole positions, and the relation does not apply for the same reason it holds elsewhere. The intraday benchmark has $\bar{w} = 1.79416$, a gap of -0.63% : the shortfall in w_{sum} is

45.73, that is, 25.5 delta-type fills at the observed average weight, vs. 27 orders missing relative to the expected total of $2 \times 2,012$. Every night of carryover removes the evening exit and turns the next morning's entry into a partial rebalance: the two counts agree to within the uncertainty on the weight of a delta fill, that is, the same explanation read from two independent sides.

3.8 What the method does not reconstruct

From the equity curve alone one cannot recover the number of orders, the win rate, the average gain and loss per trade, the expectancy, the commissions or the turnover: these are quantities that require direct instrumentation of the backtest, not its return series. The method gives the cost map, not the operating statement, which is why the sixteen configurations were run at 0 bps one by one anyway.

§4. The three levers

Given that this is a zero-beta diversifier, is it well built? The repository presents S0 to S4 as a ladder: each strategy adds something to the previous one and the return rises. This section takes the ladder apart and asks what each rung is worth on its own.

4.1 The design

S2 and S4, the two configurations on which the paper builds its final comparison, differ on **two** axes rather than one. S2 requires three confirmations at entry (band, EMA, VWAP), the third of which appears in neither published description: equations (7) and (8) of the paper (§4.2) and the README both list only the band and the EMA. It exits on a simple check every 30 minutes. S4 requires two (band, EMA) and exits on a check every 5 minutes that must be confirmed on four consecutive bars. Comparing them directly does not say which of the two changes produced the difference.

The two missing cells were built: **S2'** is S2 without the VWAP entry condition, **S4'** is S4 with that condition. A third, **S3'**, is S3 with the same condition and is used in §4.4. None is repository code, and their rows sit in a separate, labeled table.

The third axis is not a design choice but a point on which the published descriptions differ, and it is not where one would expect. The code exits a long position when $P < \max(UB, \text{vwap})$, where UB is the same upper band that authorized the entry. This is exactly what the paper specifies, in equation (5) and in the pseudocode (Code 1): **paper and implementation agree**. It is the README that diverges, describing the exit as the price crossing back through the VWAP

or through *the opposite band*, that is, $P < \max(LB, vwap)$. The two variants are called `tight`, the one specified and implemented, and `loose`, the one that exists only in the repository's prose description. Every cell exists in both.

The `tight` exit is not an idiosyncrasy of the repository: it is the dynamic trailing stop of the design the strategy descends from, Zarattini, Aziz and Barbon (2024), where the position is closed when the price crosses back through the boundary of the noise area, in the variant that takes the higher of that boundary and the intraday VWAP. What is noted here is only its geometry, which does not depend on which of the two texts one follows. A long entry occurs with $P > UB_t$, and the exit threshold at the next check is $\max(UB_\tau, vwap_\tau)$: two bands evaluated at different instants, but they share their anchor, because $\max(O_{\text{today}}, C_{\text{yesterday}})$ is fixed at the open and does not change for the rest of the day. They therefore differ only through σ , the fourteen-day average of the absolute excursion from the open at that clock minute, a quantity that grows over the session because it measures how far the price has typically already moved. In the typical case $UB_\tau \geq UB_t$, and **the tight exit threshold sits at or above the level that authorized the entry**: the stop distance is zero or negative. It can narrow locally, because σ is estimated on fourteen observations per minute and is noisy between adjacent minutes, so this is not a structural guarantee; but the gap involved remains a fraction of the band width, not a stop distance. On this reading, what the exit cadence measures under `tight` is less the whipsaw on the VWAP than the distance of the stop. How early in the life of a position the condition typically becomes true is not measured here, since the holding-time distribution was not collected (§8).

This makes `loose` more than a curiosity. The paper motivates the VWAP term by describing it as "volume-adaptive, institutionally relevant" and as a quantity that "self-widens throughout the session, giving winning trades room to run". Under the formula the paper itself writes, however, the term cannot give room to anything: when the VWAP rises above the band, the `max` selects the VWAP, which is **higher**, and the position is closed *earlier* than the band alone would close it. The term can only tighten the exit, never loosen it, and this conclusion does not depend on how often it binds. `loose` is not an arbitrary correction: it is the reading of the exit under which the published justification holds.

A note on nomenclature, because the term *vwap* appears in three distinct roles: the **entry filter** (axis 1), the **term inside the max of the exit threshold** (present in both variants, hence constant across the design), and the value `vwap` in the `exit_variant` column of the CSV files, a name inherited from the data collection that means `loose`. The text always uses `loose`.

cell	entry conditions	exit structure	in the repository
S2'	band + EMA	30' simple	no, constructed
S4	band + EMA	5' + gate $N = 4$	yes
S2	band + EMA + VWAP	30' simple	yes
S4'	band + EMA + VWAP	5' + gate $N = 4$	no, constructed

The design has **three axes**.

Axis 1: VWAP entry filter. Present or absent. It is a third condition required to open a position, in addition to the band break and the EMA filter. It moves along the pairs S2' to S2 and S4 to S4'.

Axis 2: exit structure. 30' simple (condition checked every 30 minutes, no persistence requirement) or 5' + gate $N=4$ (checked every 5 minutes, must be confirmed on four consecutive checks). It moves along the pairs S2' to S4 and S2 to S4'. The axis turns two knobs together, the **cadence** of the check and the **persistence** required, and the repository always introduces them as a pair; S1, which has the 5-minute cadence without the gate, separates them along the chain S0 to S1 to S3 (T9). It is a decomposition along a path, not a full factorial: it gives "cadence alone" and "gate given the cadence", not "gate without cadence", because four checks on a 30-minute cadence would mean two hours of persistence, a cell the repository does not contain and that would hardly make sense. The gate is a parameter of the fast-check regime, not a mechanism that can be mounted anywhere.

Axis 3: exit threshold. `tight` is the one implemented in the code, the long exits if $P < \max(UB, vwap)$; `loose` is the one described in the README, $P < \max(LB, vwap)$. The paper (eq. 5 and pseudocode) specifies `tight`, so the README is the one that differs. Every cell exists in both versions, so the cells are eight. It is the axis that makes up the rows of T8 and T10.

4.2 The VWAP entry filter is additive

Effect of the VWAP condition alone, in eight-year log return, holding the exit structure fixed:

T8. Effect of the VWAP entry filter, holding the exit structure fixed

Eight-year log return: $\ln(1 + R_A) - \ln(1 + R_B)$, that is, the logarithm of the ratio between final wealths; +0.03384 means that the cell with the filter closes with 3.4% more capital. The metric is in logs because the cells start from different levels and because only in logs does "zero interaction" mean "additive effects": in percentage points two independent effects would show a spurious interaction equal to their product, here ten times larger than the one measured. Rows: exit threshold. Columns: the level at which the exit structure is held fixed.

exit threshold	with the 30' simple exit ($S2 - S2'$)	with the 5' + gate exit ($S4' - S4$)	interaction
tight	+0.03384	+0.03339	-0.00044
loose	+0.02803	+0.02850	+0.00047

The interaction is of order $5 \cdot 10^{-4}$, that is, 1.3% of the main effect under **tight** and 1.7% under **loose**, and it **changes sign** between the two exit variants: noise, not an effect. The condition produces the same gain whether or not the S4 exit package is present, and the same under both thresholds.

One point needs clarifying at once, because it bounds the claim: all four cells of the table have the EMA filter. What is established is that the VWAP is additive **with respect to the exit structure, in the presence of the EMA**. The check in the absence of the EMA would need band + VWAP under both exit structures; only the one with the exit package was run (S3'), and the point returns in §4.4.

The ratio between how much the filter removes and how much it is worth is the interesting part. It blocks between 1.4% and 1.7% of entries, 22 to 26 trades depending on the cell out of a total of 1,450 to 1,750, and it is worth between 2.8% and 3.4% of log return over eight years. It is not a filter that improves the average quality of entries: it is a filter that intercepts a small and systematically poor subset. It is a separate source of edge, not a remedy for a weakness elsewhere in the strategy.

4.3 The exit structure: the cadence costs, the gate offsets it

The repository introduces two things together in moving from S2 to S4: the exit check goes from 30 to 5 minutes (cadence), and confirmation on four consecutive bars is required (persistence). The two appear inseparable, but the cell that separates them already exists in the repository: **S1 is S3 without the gate**, identical in everything else, with the same 30/5 intervals, the same threshold formula, no EMA filter and no VWAP filter. The paper itself describes Strategy 3 in these terms, as Strategy 1 with the persistence counter added (paper, §4.3); what it does not do is measure the two steps separately, which is what the table below does:

T9. Decomposition of the exit package (no EMA, no VWAP)

S0 = 30' simple; S1 = 30/5 without gate; S3 = 30/5 with gate $N = 4$. S1 and S3 differ only in `exit_confirmation_bars`, so cadence and persistence **are separable**.

exit threshold	cadence 30' to 5' ($S1 - S0$)	gate $N = 4$ ($S3 - S1$)	package ($S3 - S0$)
tight	-0.06508	+0.23821	+0.17314
loose	-0.04853	+0.09771	+0.04918

Orders: S0 3,594, S1 4,900, S3 3,550 (**tight**).

The cadence is costly under both thresholds. Moving from a check every 30 minutes to one every 5 brings final wealth to 93.7% of that of S0 under `tight` and 95.3% under `loose`, that is, -0.06508 and -0.04853 in log return, which are the values reported in the table, and brings the order count from 3,594 to 4,900. This is the effect the paper calls the Momentum Paradox, and it holds under both exit thresholds: it does not depend on the zero-distance threshold of `tight`, so it is not an artifact of the threshold.

The gate offsets exactly that cost. Under `loose` the cadence removes 0.049 and the gate returns 0.098: the net balance of the package is $+0.049$, so the gate recovers the cost and adds a quantity of the same order. Under `tight` the recovery is much larger ($+0.238$). There the exit threshold typically starts at, or above, the level that authorized the entry (\$4.1): a long enters when the price crosses the upper band, and the same band, which widens during the session, is the level it must stay above. Without persistence, the first pullback below it closes the position; with four consecutive confirmations required, the position survives pullbacks shorter than twenty minutes. The gate is doing the work that a stop placed some distance below the entry would otherwise do.

The conclusion to take away is not that the gate is worth zero, but **that it cannot be judged apart from the cadence**: part of what it returns only compensates for a cadence choice made two strategies earlier. The repository presents S4 as an improvement on S2; on this reading, what S4 does is pay a cost (the 5-minute cadence) and then buy its remedy (the gate), which returns more than the cost under both thresholds.

4.4 The EMA substitutes for the exit package, the VWAP does not

The same package measured in the presence of the other two entry filters:

T10. Effect of the **exit structure**, 30' simple to 5' + gate $N = 4$

Eight-year log return, first figure `tight` and second `loose`. The columns are the four combinations of the two entry filters; the fourth would need band + VWAP with the 30' simple exit, which was not run (S3' supplies that combination with the exit package only).

value of the exit package	no EMA, no VWAP (S3 – S0)	EMA, no VWAP (S4 – S2')	EMA and VWAP (S4' – S2)	no EMA, VWAP
<code>tight</code>	+0.17314	+0.12709	+0.12665	not run
<code>loose</code>	+0.04918	-0.00344	-0.00297	not run

Possible readings, each at a **single level** of the other filter:

- **effect of the EMA on the package** (col. 2 minus col. 1), measured with the VWAP absent: -0.04604 (tight) and -0.05262 (loose). Negative interaction, so the EMA and the package are **substitutes**
- **effect of the VWAP on the package** (col. 3 minus col. 2), measured with the EMA present: -0.00044 and +0.00047. Zero interaction, so **additive**
- **effect of the EMA on the VWAP**, with the exit package in place: the VWAP is worth +0.00804 without the EMA ($S3' - S3$) and +0.03339 with it ($S4' - S4$) under tight, +0.00650 and +0.02850 under loose; interaction +0.02536 and +0.02200. Positive, so the two entry filters are **complements**. $S4'$ beats $S3'$ by +0.04526 and +0.03000

The second and third columns are identical to each other and different from the first. That is: **adding the VWAP filter does not change the value of the exit package; adding the EMA filter reduces it by about 0.05, from 0.173 to 0.127 under tight and from 0.049 to about zero under loose**. The $\text{EMA} \times \text{package}$ interaction is -0.046 under tight and -0.053 under loose.

The paper does not leave this point implicit. In the section devoted to $S4$ it states that the two mechanisms "are structurally independent: the EMA filter operates at the entry decision gate, while the persistence counter operates on the exit signal after a position is open", and that their combination is therefore "not merely additive but potentially synergistic". From there it draws its formal hypothesis:

Hypothesis 1. *If Q_{entry} and Q_{exit} are independently improvable, a mechanism that maximises both simultaneously dominates any mechanism that maximises only one.*

The two parts should be kept separate. The **conclusion** holds: $S4$ does dominate $S2$ and $S3$, it beats both on return and on Sharpe, and it remains the best configuration in the repository. It is the **premise** that the table above measures, and the data do not support it. If the two mechanisms were independent the exit package would be worth the same with and without the EMA; if they were synergistic it would be worth more. It is worth less in both variants, by an amount, -0.046 and -0.053 , that is two orders of magnitude above the interaction noise measured in §4.2 on the same scale. They are not independent: they are partially redundant.

What does not hold, then, is not the paper's result but the reason given for it, and with it the principle that the $S0$ to $S4$ ladder presupposes, namely that stacking mechanisms is the way forward. The paper writes "*potentially synergistic*", and that caution should be acknowledged: the statement was conditional. It has been tested here, and the measured interaction points the other way.

The EMA filter and the exit package are **substitutes**: both deal with breakouts that do not continue, one by keeping them out and the other by not closing on their noise, and having both does not pay twice. The VWAP condition does not overlap with the package: its value is the same under both exit structures (T8).

Hence the answer to the opening question. Under the specified and implemented threshold, the best of the sixteen is S4': it is S4 plus the one entry condition that does not overlap with the exit package. The lead is not uniform across the two axes, however: under `loose` the exit package is worth zero and S2, which is repository code, is on a par with S4' (259.7 vs. 258.6 at 0 bps). The full comparison is in §6.

One combination of the two entry filters was missing from the design, `band + VWAP without EMA`. It was run with the exit package, as S3', to check whether S4' is beaten once the EMA is dropped. It is not: S4' beats it by 0.045 in log wealth under `tight` and 0.030 under `loose`. The run also shows that the two entry filters are not independent: with the exit package in place, the VWAP is worth 0.0080 without the EMA and 0.0334 with it (0.0065 and 0.0285 under `loose`), a positive interaction, so they are **complements** (T10). The same combination with the 30' simple exit was not run, so each conclusion of the table above still holds at one level of the other filter only (§8).

4.5 How much the zero-distance threshold weighs

A prediction of this analysis that the data did not bear out, reported as such.

Under `tight` the exit threshold sits at or above the level that authorized the entry, so the condition should become true early in the life of most positions: the cadence would then stop measuring the readiness to react to a reversal and measure instead how long a position the threshold has already condemned is allowed to run, half an hour vs. five minutes. If the gap between S0 and S1 were entirely due to this, then under `loose`, where the threshold is genuinely distant and fires only on a real retracement, the gap should close.

It does not close: it goes from 0.06508 to 0.04853, so **the zero-distance threshold explains 25% of it** and the rest is genuine. The 5-minute monitoring really is costly, and the paper's qualitative conclusion survives the threshold, even against the alternative this analysis was testing.

The paper does not stop at recording the phenomenon, however: it gives it a mechanism. *"VWAP is a slow-moving quantity that generates brief, transient violations of the composite exit condition with no predictive value [...] Strategy 1 frequently closes valid positions on these transient violations, the classic whipsaw outcome."* On the paper's account, the cost comes from the VWAP term.

This analysis cannot put that to the test, and the reason matters, because the check seems within reach and is not. The exit condition is $P < \max(\cdot, \text{vwap})$, where the first term is UB under `tight` and LB under `loose`; LB is anchored below the open while the VWAP is an average of the prices traded during the session, so for a long position the maximum under `loose` is the VWAP in the great majority of cases. One would then be tempted to read the smaller cost of S1 under `loose`, 0.04853 vs. 0.06508, as evidence against the stated mechanism: the cost is smaller precisely where the VWAP is in charge.

It is not such evidence. The two variants differ not only in *which* term binds but also in *how far* the threshold is from the entry level, and a more distant threshold is crossed less often whatever the quantity that defines it. The smaller cost under `loose` is what one would predict from the distance alone, without ever mentioning the VWAP: the two effects push in the same direction and cannot be separated with these cells. Add that under `tight` the VWAP is not absent (when, on a day of sustained rally, it rises above UB , it binds there too), and that the paper's mechanism concerns the *rate* of transient violations while what is measured here is the *total* cost.

What the data establish is therefore the fact and not the mechanism: the five-minute cadence has a cost, and it has a cost even where the threshold is at a real distance. Attributing that cost to the VWAP rather than to the denser sampling of a noisy series would require a cell the repository does not contain, namely the same cadence with an exit condition free of the VWAP term. It is the one statement of the paper about S1 that this report leaves standing without being able to either confirm or rule out.

On the ordering between the two variants the picture reverses as we move from S0 to S2 to S3 and S4. On S0, S1 and S2, `loose` returns more than `tight` already at 0 bps. On S3 and S4 it returns less, but it has a lower w_{sum} and therefore decays more slowly as costs rise:

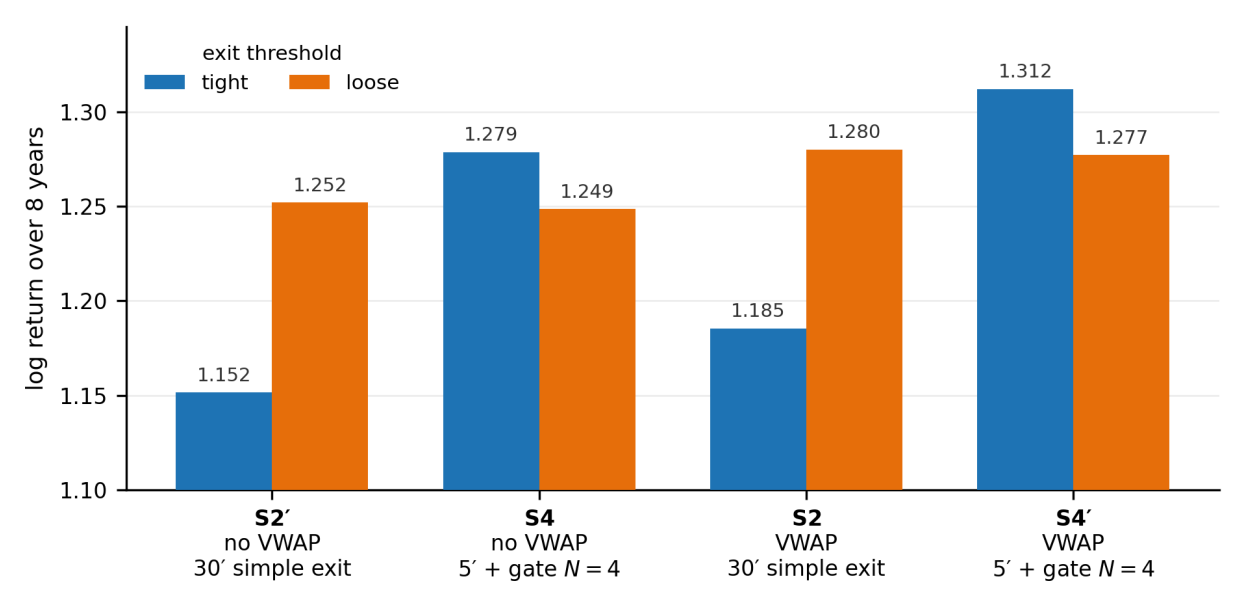
T11. Crossover between the two **exit thresholds** (axis 3)

Slippage level at which `tight` and `loose` break even, for a fixed strategy.

strategy	<code>tight</code> at 0 bps	<code>loose</code> at 0 bps	w_{sum} <code>tight</code>	w_{sum} <code>loose</code>	crossover (bps)
S0	196.12	229.19	6,500.9	5,654.9	-1.251
S1	177.47	213.60	8,873.8	7,064.1	-0.676
S2	227.20	259.71	6,228.1	5,397.6	-1.141
S3	252.10	245.79	6,430.5	5,566.4	0.209
S4	259.18	248.56	6,123.0	5,283.1	0.357

Negative crossover = **loose** already wins at 0 bps and the gap widens. On S3 and S4 the sign reverses at 0 bps, but **loose** has a lower w_{sum} and therefore decays more slowly. Whether the overtake falls at a realistic level of slippage is discussed below.

The crossover falls at **0.209 bps** on S3 and **0.357 bps** on S4. Since the 0 bps runs already pay the quoted bid-ask spread (\$1.1), these are costs on top of it: loose overtakes tight on S3 and S4 only if execution is worse than the quote by more than that. This report does not measure how large that excess is, so it cannot say which of the two exits wins once it is included (§8).



F2. The four cells of the VWAP entry filter × exit structure plane, all with the EMA filter, under the two exit thresholds

§5. Ranking stability

The question is whether the conclusions of the previous sections survive when slippage is added, and the answer is that they survive almost entirely: the ranking of the repository's five strategies is stable within each exit variant across all five cost levels collected, both by return and by Sharpe. This is a result in the paper's favor, and it deserves recording as such. There is a single exception, and it is worth setting out.

T12. Ranking stability across the five cost levels

variant	metric	stable	ranking
tight	return	yes	S4 > S3 > S2 > S0 > S1
tight	Sharpe	no	0.00: S4 > S3 > S2 > S1 > S0 · 0.25: S4 > S3 > S2 > S0 > S1 · 0.50: S4 > S3 > S2 > S0 > S1 · 1.00: S4 > S3 > S2 > S0 > S1 · 2.00: S4 > S3 > S2 > S0 > S1
loose	return	yes	S2 > S4 > S3 > S0 > S1

variant	metric	stable	ranking
loose	Sharpe	yes	S2 > S4 > S3 > S0 > S1

The single inversion (S1 vs. S0 on Sharpe, **tight**, between 0 and 0.25 bps) is discussed below. The hypothesis "turnover, not the number of trades, predicts the drag" cannot be tested on these data: by the relation of T7 the two are almost proportional, so there is almost no variance to explain.

Under **tight**, the ranking by return is $S4 > S3 > S2 > S0 > S1$ at every level, from 0 to 2 bps. Under **loose** it is $S2 > S4 > S3 > S0 > S1$, again unchanged. For Sharpe the same holds across the whole table except at one point: at 0 bps, under **tight**, S1 has a Sharpe of 0.841 vs. 0.835 for S0 (0.842 vs. 0.835 in the authors' file), and therefore sits above the baseline. At 0.25 bps of additional slippage the order reverses, 0.585 vs. 0.665, and it does not reverse back at any of the subsequent levels. It is the only inversion in the whole eight-year table.

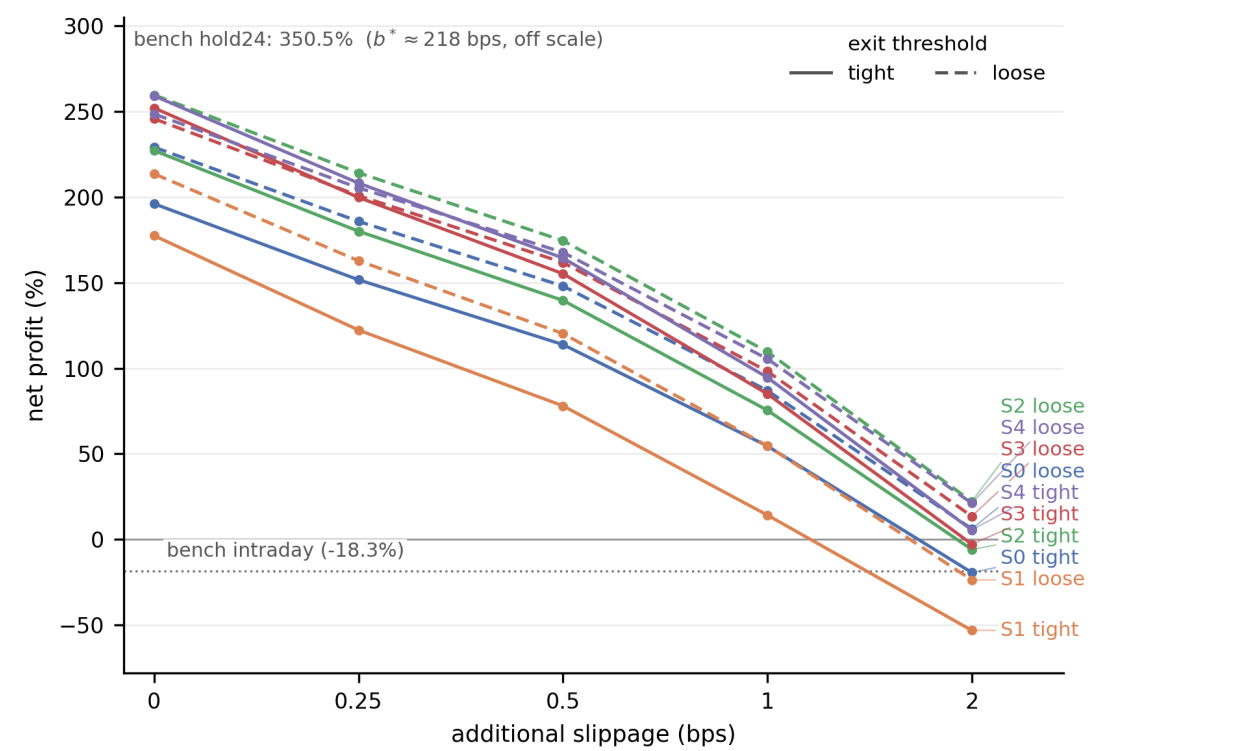
It concerns the comparison, S0 vs. S1, on which the paper builds the Momentum Paradox, and it is not a point against the paper. The authors make no claim on that step of Sharpe: their sentence is "*degrading total return (196% $\rightarrow$ 177%), Sharpe (0.835 $\rightarrow$ 0.84, a near wash), and win rate (40% $\rightarrow$ 32%)*", and the parenthesis is theirs. They are right not to lean on it: +0.005 separates two configurations that share most of their trades, and it reverses at the first cost level, as the next paragraph shows. The Momentum Paradox does not rest on that number anyway but on the return, which loses nineteen points and never moves in the ranking: S1 remains last at all five cost levels.

What costs add is that the one ambiguous ordering stops being ambiguous. At 0.25 bps the Sharpe of S1 is no longer a wash but a clear deterioration, and all four metrics of that step move in the stated direction. **Costs do not overturn the paper's conclusion: they sharpen it.** It is the case in which the strategy most penalized by costs is also the one the paper identified as the weakest, and the coincidence is not accidental: S1 has 4,900 orders vs. the baseline's 3,594, that is, the highest w_{sum} of the sixteen configurations, so it is the row that decays fastest as soon as the cost rises.

5.1 The same standard, applied to the paper's own ranking

The paper breaks its trades down by weekday and reports the ordering Friday > Wednesday > Thursday > Tuesday > Monday, reading it as "a robust feature of market microstructure rather than a mislead caused by overfitting" (paper, §5.5, Table 9). The selection question this report applies to itself in §A.3 — sixteen configurations searched, one reported — applies to any ranking over five groups. Applying it to one's own results and not to the ranking one is checking would make the severity selective, so it is applied here as well.

The authors' own files answer it. Aggregating DayOfTheWeek/Trades_Strat{2,3,4}_8y.csv by entry date, with P&L net of the fee column, so that the several trades of one day count once and within-day correlation cannot inflate anything, and testing each weekday's mean session return against zero:



F3. Decay under costs. The 0 bps baseline already includes IB commissions

Day-of-the-week on S4: mean session return by weekday

	Mon	Tue	Wed	Thu	Fri
mean session return	+0.043%	+0.071%	+0.157%	+0.088%	+0.190%
<i>t</i>	0.77	1.26	2.48	1.57	3.31
sessions	214	235	233	271	244
<i>p</i>	0.444	0.208	0.014	0.117	0.00109

Sessions with at least one trade, 1,197 in total; one-sample *t*-test of the mean against zero, two-sided. The same computation on S2 and S3 returns the same ordering, with Friday at *p* = 0.0013 and *p* = 0.0007 respectively.

The ordering reproduces the paper's exactly. Friday's mean carries *p* = 0.0011 on its own and 0.0054 once multiplied by the five groups the maximum was selected from, so it stands at the 1% level under the most conservative correction available; Monday is indistinguishable from zero (*p* = 0.44), consistent with the paper's reading of it as the weakest session. The ranking survives the standard this report applies to itself, and that is a result in the paper's favor.

One check falls out of the same reconstruction at no cost. Compounding those session returns from the trade files alone, with no backtest involved, returns cumulative eight-year figures of 227.331%, 252.214% and 259.235% for S2, S3 and S4, against the 227.320%, 252.248% and 259.235% of the authors' result files: within four hundredths of a point on the first two and coinciding to the third decimal on the third. It is an independent verification of three rows of T1, reached from the trade lists rather than from a re-run.

§6. S4': a configuration the repository does not include

§4 decomposes the repository's design and finds that the VWAP entry filter adds edge that the exit package leaves intact and that, with the package in place, is larger when the EMA filter is present. If so, the best configuration is not S4: it is S4 with that condition added, S4', and it was never run in the repository because no cell crosses all three entry filters with the persistence-based exit structure at the same time.

T13. S4' vs. the best configurations in the repository

	0 bps	0.25 bps	0.5 bps	b^* (bps)	drawdown	Sharpe
S4' tight	271.4	219.4	174.8	2.178	9.0%	1.075
S4' loose	258.6	215.0	176.7	2.462	8.9%	0.970
S2 loose	259.7	214.2	174.6	2.372	9.6%	0.999
S4 tight	259.2	208.2	164.4	2.088	8.4%	1.036

The 0.25 and 0.5 bps columns of S4' are **predicted** by the model of T6, not measured: for the constructed cells only the 0 bps run exists. The lead of S4' tight over S4 tight is +0.03339 in log return over eight years (T8), that is, 3.4% more final capital, about 0.4% per year.

At 0 and at 0.25 bps the leading configuration is S4' tight. At 0.5 bps S4' loose takes the lead, since it has a lower w_{sum} and therefore decays more slowly, and the top three close to within two percentage points: 176.7 vs. 174.8 for S4' tight and 174.6 for S2 loose. In one form or the other S4' is therefore first at all three levels, but it is not always the same variant, and the highest break-even of the sixteen configurations does not belong to S4' tight (2.178 bps) but to S4' loose: **2.462 bps**.

Two questions follow, and they are separate. Among the cells that are in the repository, the best at 0 bps is S2 loose, not S4 tight: 259.7% vs. 259.2%, and 2.372 bps of break-even vs. 2.088. Completing the factorial then adds **11.7 percentage points of return** over that best repository cell, 271.4% for S4' tight against 259.7%, and **0.09 bps of cost tolerance**, 2.462 for

S4' loose against 2.372. The second figure is small in absolute terms but it is 3.8% of the available margin, on a family whose break-evens all lie between 1.15 and 2.46 bps.

The size of the lead should be kept in view. Over S4 tight, the paper's own final configuration, S4' tight gains +0.033 in log return over eight years (T8), that is, 3.4% more final capital, about 0.4% per year, and the whole of it comes from the 22 to 26 trades the entry condition blocks (§4.2). One caveat cuts against the result, because the table shows it: **S4' does not dominate S4 in the Pareto sense**. Return, Sharpe and break-even improve; the drawdown worsens, 9.0% vs. 8.4%. The dominance the paper reports for S4 cannot be repeated for S4' here. The Deflated Sharpe Ratio of Appendix A, which finds S4' distinguishable from selection noise, tests it against a null of zero edge, not against S4: whether the margin between the two survives outside this sample is a question these data do not answer (§8). S4' is best read as S4 plus a small increment of consistent sign, not as a different strategy.

That the 0.25 and 0.5 bps columns of the S4' rows are predicted by the model of §3 rather than measured is consistent with the rest of the method, since the cost model was validated on forty independent rows; but it is the only part of the table that does not come from a direct backtest, and §8 records it as such.

The register in which this result is presented matters as much as the result itself. It is not "this analysis beat the repository": the condition that separates S4' from S4 is already written in the repository, in S2, and crossing it with the exit structure of S4 required no new idea, only a cell of the factorial that had not been run. This is why S2', S3' and S4' stay in a separate table and are labeled as constructions of this report rather than as repository code. The distinction between "what the repository measured" and "what the factorial implies" is the point of the section, not a formatting detail.

§7. Reconciling the paper with the code

Four items in which the published description, the published code and what reproduces today do not coincide. Two lie in the text of the paper, one concerns the reproducibility of a platform statistic, and one is a detail of the committed source, stated for completeness.

1. The Probabilistic Sharpe Ratio does not reproduce. The paper reports a PSR of 83.4% for S4, which is exactly what the authors' own stats file contains (83.439%). Re-running the same code today returns 44.342%, and this is not an isolated case: the same difference, 39 to 46 points, appears on all five strategies (T1). The paper transcribes its own output correctly, and this is the only platform statistic that does not come back after a lapse of time on the same code, since orders, Sharpe, drawdown, win rate and turnover all agree to within rounding.

What changed is the platform, not the original work, and the change can be identified. LEAN's own routine, `Statistics.ProbabilisticSharpeRatio`, with the benchmark Sharpe of 1 that Appendix A, §A.8 documents, reproduces both columns once a single detail is allowed to differ between the two dates: the risk-free rate inside the Sharpe ratio that the PSR tests. Without it, the routine returns the authors' values (61.7% against 61.6% on S0, 83.2% against 83.4% on S4); with the platform's rate of about 2.7% subtracted, it returns today's (21.5% against 21.5%, 44.4% against 44.3%), to within 0.3 points on all ten figures. The printed Sharpe ratio subtracts the rate on both dates, which is why it reproduces. The check starts from each run's printed Sharpe and annual standard deviation and uses, for all five strategies, the skewness and kurtosis measured on S4' tight, since they are not reported per strategy; the table is in the supplementary file.

The two dates therefore report two different statistics under the same name: in March the PSR tested the Sharpe ratio before the risk-free rate, today after it. Each is well defined, but they cannot be compared with each other, and neither is the probability that the true Sharpe ratio is positive, since both test against a Sharpe of 1. This is why Appendix A computes the Deflated Sharpe Ratio from the raw moments of the returns rather than from the platform's figure.

Provenance of the two sides of this comparison. An argument that a platform has moved is only worth as much as the two runs behind it can be dated, so both are recorded here. On the authors' side the state block of each `stats/strat{s}_8y.json` carries the backtest's own timestamps: S1, S2 and S3 ran on 18 March 2026 between 14:47 and 14:59 UTC, S0 on 20 March at 17:32 and S4 on 23 March at 08:34, on nodes BACKTESTING-80 and BACKTESTING-219, and the order counts in those same blocks are the ones reproduced in T1. Those files record no engine version, so the build the authors ran on is not recoverable from them. On this side, the reproduction used the repository at commit `e60646cd7c05b94b59f555122150dd1aa19ba5e0` and ran between 26 August and 3 September 2026. The strategy sources have not changed since commit `b960d36` of 23 March 2026: the two commits that follow it add the paper and the trade files of `DayOfTheWeek/` and touch no code, so the `strats/` directory of the commit used and of the authors' final state are the same files. What is not fixed is the engine. The projects are configured to follow QuantConnect's master branch, so each backtest ran on whichever build was deployed that day: the S0 runs of 26 August report LEAN engine v2.5.0.0.18034 in their log header, and master stood at v18057 on 4 September, twenty-three builds later in nine days. Five months of that cadence separate the two columns of the table above. This dates the gap; which build changed the definition is not identified here.

2. S2 requires three entry conditions, not two. Equations (7) and (8) of the paper describe a "dual confirmation" entry, $P_t > UB_t$ and $P_t > EMA_{100}(t)$. In `strategy2.py` the entry requires

three: band and VWAP and EMA. The difference is in the paper's own text, not only in the README that summarizes it, and this is the one item of the four that touches the interpretation of the results: the S2 vs. S4 comparison, half of the final comparison in Table 7, is confounded on two axes rather than one. It is the reason S2' and S4' were constructed (§4, §6).

3. The 15:58 liquidation does not fire on half-day sessions. The paper states that all positions are compulsorily liquidated at 15:58 ET. The branch `hour == 15 and minute >= 58` is never reached on the seventeen half-day sessions of the window, which close at 13:00, and the position stays open overnight. The magnitude is negligible, but the statement does not hold as written on those days. The seventeen dates can be verified one by one against the exchange calendar, and they account for **17 of the 27 carryover nights** identified by an entirely independent route in §3, from the w_{sum} shortfall on the benchmark. The other ten have no verified explanation in this report, and it is more accurate to say so than to round the counts until they agree.

4. The EMA warm-up is shorter than the indicator. The 100-period EMA, described as a structural trend-confirmation filter, is built with `Resolution.Minute` and 100 periods, while the warm-up is `SetWarmUp(14, Resolution.Daily)`: 14 daily bars, not 100 one-minute bars. As a result `ema.IsReady` becomes true only about 86 minutes into the first usable session, not before the start as the declared warm-up would suggest. Anyone who opens the code can verify this, and it is stated here for completeness.

§8. Limitations

This report does not establish that S4' holds up over time. All sixteen configurations are measured on a single window, 2017-05-10 to 2025-05-10, and the best of the sixteen emerges from a condition, the VWAP entry filter, that blocks only 1.4% to 1.7% of entries. That is exactly the case in which the full sample can mislead, because an advantage concentrated on a few dozen trades is more sensitive to the sub-window on which it is measured than an advantage spread over three thousand. No rolling-window or out-of-sample analysis over time was carried out. The Deflated Sharpe Ratio of Appendix A answers a different question, and answers it well (whether the best of sixteen is distinguishable from selection noise), but it does not answer this one: it is not a temporal validation, and it is declared as such in its own limitations. This is not a gap that separates this report from the paper, since the "subperiod analysis on rolling 2-year windows to detect strategy decay" appears among the paper's own lines of future work, §6.7; but sharing an open question is not the same as having closed it. It remains the check that both lack, and it is the first thing to do.

A second gap lies in the design. Band + VWAP **without** EMA was run only with the exit package (S3', §4.4), not with the 30' simple exit, so each of the two conclusions of §4.4 holds at one level of the other filter only, and the three-way interaction is not measurable.

One more cell is missing, and its absence prevents attributing a mechanism rather than an effect. To establish whether the cost of the five-minute cadence comes from the VWAP term, as the paper suggests, or from the denser sampling of a noisy series, one would need the same cadence with an exit condition free of the VWAP. The repository does not contain it, and the two available variants do not stand in for it because they also differ in the distance of the threshold: §4.5 therefore establishes that the cadence has a cost, not why it does. For the same reason the statement in §4.1 that the `tight` threshold typically binds early in the life of a position is a geometric argument, not a measurement: the holding-time distribution of the trades was not collected.

Half of the measured cells describe an exit that no formal specification defines. The `loose` variant is not the paper's exit (§4.1 shows that on `tight` paper and code agree) but a reading of the README's prose description. The reasons for measuring it are good, since it is the variant under which the VWAP genuinely binds the exit and therefore the only one in which the published justification of that term can be checked; but the results that concern it (the crossover, S4' `loose`, half of §4.5) need to be understood as counterfactuals on a configuration that exists only as a reading of the README's prose, not as measurements of what the repository does.

The cost reconstruction of §3 holds for this specific code architecture, in which entries and exits read `data[symbol].close` and not a fill price. If a future version of the signal read actual execution prices, the invariance of the trade sequence with respect to slippage would fall, and with it the whole one-backtest method.

The cost model itself is linear in b and uniform across all fills, regardless of time of day, order size or the volatility regime of the moment. It models neither market impact nor any dependence on intraday liquidity, which for the same notional traded could weigh differently on an entry at 9:31 and one at 15:55. The level of additional slippage a real execution would incur is not measured either. The 0 bps runs already pay the quoted bid-ask spread, so what the cost axis adds is execution worse than the quote; the 0.25 to 0.5 bps used in §6 is a working assumption for it, not a calibrated figure. The ranking of `tight` and `loose` on S3 and S4 depends on it (§4.5); the break-even levels do not.

One cost is not in the model at all, and it is worth saying why it need not be. The strategy is short about half the time it is in the market: on S4, 816 of its 1,694 trades are short (48.2%), and they account for 45.7% of the time at market, counting the positions that open and close

in the same session, with a mean holding period of 112 minutes against 124 on the long side. Stock borrow, however, accrues on short positions held at the close of business, and the schedule, flat every evening, keeps the strategy out of that charge almost entirely. Over the eight years exactly five positions survive to the next session, all of them opened on half-day sessions where the 15:58 liquidation branch is never reached (§7, item 3), and **only one of the five is short**: 913 shares of SPY carried over the Christmas holiday of 2018, a notional of 194,000 dollars held for two nights. On an instrument that is general collateral the charge on that one position is a few dollars over the whole window — 2.69 at a borrow rate of 0.25% annualized, 10.78 at 1% — so the item is below the resolution of every comparison in this report. The same fact disposes of a second one: a strategy that is short on 48% of its trades but never over a weekend carries no exposure to recall or to a borrow rate that moves, which is what makes short exposure costly in practice on names that are not general collateral. This is a property of the schedule, not a result of the analysis, and it is recorded here because the cost sections would otherwise appear to have overlooked it.

Two limits concern the result on the opening half-hour in §2, and they deserve stating alongside it. The first: the gap between entry at 9:31 and entry at 10:00 is measured, 17.8 points, but it is a net figure, not a decomposition. The extra half-hour brings together the drift of the window and the additional variance that a leveraged position pays as drag, and the two terms have opposite signs. Separating them was attempted in two ways, by difference between the two matched runs and with a run on the opening half-hour alone, and the two estimates do not agree with each other: the gap is of the order of one round trip of execution cost, which cancels in the matched comparison and does not in the isolated run. The text therefore reports the net figure and not its components. The second: the two runs that produce it both run at zero additional slippage, so they pay the per-share commission and the quoted spread, and nothing for execution worse than the quote. Near the open that extra cost is plausibly larger than at 10:00, so the run matched to the open may receive a discount it would not have in practice, and the 17.8 points are best read as an upper bound. The cost model just described does not correct for this, because it applies a single b to all fills and has no time-of-day dimension; switching it on, the two runs would worsen by almost the same amount and the gap would stay where it is.

The distance to be covered can be bounded, however. Since the 15:58 exit is common to both runs and cancels, the cost has to be loaded on the entries alone, whose w_{sum} is half of the total: for the gap to vanish, an extra cost of **0.684 bps** on every 9:31 entry would be needed, paid every day for eight years. The threshold is kept in basis points because that is where it is invariant: converted to cents it has no single value, since SPY grew a great deal over the sample while the tick stayed at one cent. The same threshold is worth fewer cents in 2017 than in 2025, and the constraint, if it binds, binds at the start of the sample. It is nonetheless a

wide threshold for an instrument on which one cent covers the whole quote for most of the session, and this is why the result of \$2 is reported rather than withdrawn. How much worse than the quote execution at the open actually is, however, is not measured here; it is a judgment about the instrument, not a datum of this report.

The report covers a single instrument (SPY), a single time window and a single rate regime. The risk-free rate $r_f = 2\%$ and the financing rate of 3.5% are both conventions declared explicitly, not calibrations. Neither is load-bearing: the Deflated Sharpe Ratio moves only between 99.6% and 98.0% for risk-free rates between 1% and 3% (Appendix A, §A.4.3), and the financing rate is a common-sense choice over a reported sensitivity range (T5), not a measured value.

Two clarifications on the provenance of the numbers. The 0.25 and 0.5 bps columns of the constructed cells are predicted and not measured (§6): they are the only part of the report that does not come from a direct backtest. And the reproduction took place on a platform that has moved in the meantime: relative to the authors' backtest files, commissions come out higher by between 0.24% and 0.51% and the return lower by up to 0.29pp, in addition to the PSR gap discussed in §7. These are small differences of constant sign, but they rule out treating the last digits of any comparison as exact.

Finally, the Deflated Sharpe Ratio corrects for selection **within** the family of sixteen configurations, not for the choice to study this family in the first place: that selection, the publication of a strategy because it worked, took place upstream, and no downstream correction can recover it. It is a structural limit of any analysis built on someone else's code, not specific to this report, but it needs saying, because otherwise the DSR risks being read as a stronger guarantee than it is.

Appendix A. Deflated Sharpe Ratio on S4' tight

Methodological appendix to the independent analysis of `blackswan-quants/intraday-momentum`

A.1 The problem

One result of this report is that the best configuration among those examined, S4', that is, S4 with the VWAP entry filter added, reaches a Sharpe of 1.075 vs. 1.036 for S4 and 0.835 for the S0 baseline.

This result was obtained **by selecting the maximum over sixteen configurations**. Selection introduces a bias that has nothing to do with the quality of the strategy: the maximum of N noisy estimates is systematically above the mean of the population they come from, even when that population has a mean of exactly zero.

The most direct analogy: sixteen perfectly fair coins, a hundred tosses each, and one keeps the coin with the most heads. That coin will show a frequency of heads above 50%. Not because it is biased, but because it is the maximum of sixteen draws.

The right question is therefore not "*is 1.075 a high Sharpe?*" but:

How high would the best of the sixteen have been if none of the sixteen had any edge?

This appendix computes that threshold and compares the observed result against it.

A.2 Why the PSR reported by QuantConnect is not enough

For every backtest QuantConnect reports the **Probabilistic Sharpe Ratio** (Bailey and López de Prado, 2012), which addresses a different and equally real problem: the sample Sharpe is an estimate, and its uncertainty depends on the length of the series, on skewness and on the thickness of the tails. The PSR measures how far the sample Sharpe stands above a fixed threshold, in units of its own sampling error:

$$\text{PSR}(SR^*) = \Phi \left[\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right]$$

where $\hat{\gamma}_3$ and $\hat{\gamma}_4$ are the skewness and the (non-excess) kurtosis of the returns, n the number of observations and Φ the standard normal CDF.

One point on how this quantity is to be read, because it propagates to everything below. The expression standardizes the distance between an estimate and a threshold and passes it through Φ , so the number it returns is at once the complement of a one-sided p-value and, reading $\widehat{SR}$ as a pivotal statistic for a location parameter, the confidence that SR exceeds SR^* . The two readings agree for as long as SR^* is a constant fixed in advance, which is the case for the PSR. They stop agreeing once the threshold is itself estimated from the same data, which in the deflated version used here happens only through a plug-in worth a hundredth of a point (\$A.10). The figure of \$A.6.2 is therefore reported as a test statistic with a nominal level, not as a probability attached to the parameter.

The limitation is that the PSR looks at **one** strategy at a time and does not know how many were tried. Applying it to all sixteen and reporting the best means reporting the maximum of sixteen PSRs, that is, falling back into the original problem.

The **Deflated Sharpe Ratio** (Bailey and López de Prado, 2014) is the same formula with the threshold changed: instead of $SR^* = 0$ one uses SR_0 , the maximum Sharpe expected under the null hypothesis H_0 , which from here on always denotes the following: **all** the configurations tried have exactly zero edge.

$$DSR = PSR(SR_0)$$

A.2.1 A second reason, independent of the first: the platform PSR does not reproduce

There is a second reason, distinct from multiple selection, not to build the DSR on the PSR that QuantConnect reports in its interface: that number has not remained the same over time on the same code, because its definition has changed.

The repository preserves `stats/strat{0..4}_8y.json`, the authors' backtest output. Re-running the same code without modifications, over the same period, five months later — their runs are of March 2026, these of August and September 2026, and §7 records the provenance of both:

	PSR in the authors' file	PSR re-run today	difference
S0	61.589%	21.510%	40.1 points
S1	67.736%	21.982%	45.8 points
S2	75.319%	33.461%	41.9 points
S3	81.164%	41.219%	39.9 points
S4	83.439%	44.342%	39.1 points

It is the only statistic that moves. Compared with the same file, which reports 3 to 4 significant figures rather than the rounded values of the README, orders (exact, e.g. 3,388 on S4), Sharpe (within 0.001), Sortino, drawdown, win rate, α , β , annualized standard deviation and Portfolio Turnover all agree. The strategy sources are unchanged since commit b960d36 and the order counts match exactly on all five rows, so the inputs to the computation are the same on the two dates up to the fill-price differences documented in §1, which move the returns by at most 0.29pp and cannot account for forty points of PSR. What is not fixed between the two dates is the engine build, since the projects follow QuantConnect's master branch, and §7 identifies what changed in it: LEAN's routine reproduces the March values with no risk-free rate inside the tested Sharpe ratio and today's with the rate subtracted, to

within 0.3 points on all ten figures. The build in which the change entered is not recoverable from the result files.

This is the practical reason, in addition to the statistical one of §A.2, why the DSR of this appendix does not start from the PSR reported in the interface of any backtest, neither of our own sixteen nor of the March runs preserved in the repository. It starts from the five scalar accumulators of §A.5, computed locally from the daily returns of the same run: a quantity that does not depend on which version of QuantConnect's pipeline produced it. The full detail of this comparison is in §7 (T1).

A.3 The null threshold SR_0

Under H_0 the N sample Sharpes are draws from a distribution with mean zero and standard deviation σ . The expected value of their maximum is a problem in extreme value theory: for Gaussian draws the maximum converges to a Gumbel distribution, and the closed-form approximation is

$$SR_0 = \sigma \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

with $\gamma \approx 0.5772$ the Euler-Mascheroni constant. The two terms are the quantiles corresponding to the mode and the mean of the limiting Gumbel, combined with weight γ . For $N = 16$ the quantity in brackets is 1.8005.

The relevant property is that $\Phi^{-1}(1 - 1/N)$ grows like $\sqrt{2 \ln N}$, more slowly even than a logarithm: the threshold rises very slowly as the number of trials increases.

A.4 Parameters: choice and justification

A.4.1 Number of trials, $N = 16$

The sixteen configurations for which a 0 bps backtest exists: S0, S1, S2, S3, S4, S2', S3', S4', each in the two exit variants `tight` and `loose`.

Ten of these belong to the initial design. The three prime cells (S2', S3', S4') were **constructed along the way**: S2' and S4' after observing the results of S2 and S4, to separate two axes that were confounded in the original comparison, entry filter and exit structure; S3' after observing the interactions of §4.4, to run the one combination of the two entry filters missing under the exit package. This is data-driven research, not a design fixed in advance, and it is

declared as such: the motivation was one of experimental design, not of expected result, but the cells were added after looking at the numbers.

No configuration was explored and discarded: the sixteen are the entire search space and all are reported. The variants at $b > 0$ do not count as separate trials, because they are not independent searches but deterministic transformations of the 0 bps rows through the cost reconstruction.

$N = 16$ is also **conservative** with respect to the independence assumption: the trials share instrument, window, leverage rule and most of the logic, so the number of effectively independent trials is below 16, which would lower SR_0 .

A.4.2 Null dispersion, σ : two specifications

Main specification (conservative). Under H_0 the relevant dispersion is the standard error of the sample Sharpe of one independent trial whose true Sharpe is zero, $1/\sqrt{n} = 0.02230$. The main specification plugs the observed Sharpe into the general expression instead,

$$SE(\widehat{SR}_d) \approx \sqrt{\frac{1 + \widehat{SR}_d^2/2}{n}} = 0.02234$$

which is 0.2% larger and therefore raises the threshold slightly. Both are already in daily units; the conclusions rest on the second, and §A.6.2 reports the first alongside it.

Bailey and López de Prado specification. The authors prescribe the **cross-sectional** standard deviation of the N sample Sharpes; over the sixteen values at 0 bps it is $\sigma_{\text{cross}} = 0.06795$ in QC Sharpe units.

Here this specification is **not informative**, for a reason that matters: the sixteen trials are minor variants of one another, so they produce nearly identical Sharpes. σ_{cross} measures the variability *between variants*, not the sampling variability that an independent trial would have under the null hypothesis, and it is in fact smaller than the theoretical standard error by a **factor of 4.33**: 0.06795 vs. $0.02234 \times 13.173 = 0.29430$, both in QC units. The resulting figure is so high as to be insensitive to any N , which is in itself the symptom that the threshold is too low. It is reported in §A.7 for completeness, not as a result.

Two details on the threshold:

- Under H_0 the higher moments do not enter the SE. The non-normal expression, $\sqrt{(1 - \hat{\gamma}_3 SR + \frac{\hat{\gamma}_4 - 1}{4} SR^2)/n}$, reduces to $1/\sqrt{n}$ when the true Sharpe is zero, so the skewness and kurtosis of §A.6.1 act on the denominator of the test, not on the threshold. Evaluated at $\widehat{SR}_d$ it would give 0.02055 and a lower threshold, but that is not a null quantity and it is not used.
- σ_{cross} is computed over the sixteen Sharpes **including the maximum**, which slightly inflates the dispersion and therefore the threshold.

A.4.3 Risk-free rate, $r_f = 2\%$

A convention already declared elsewhere in the report, applied as $r_f/252$ per trading day. It is not calibrated on anything: it is fixed a priori, and it is ours, not the platform's, whose own rate over this window is about 2.7% (§A.8). Its effect on the result is bounded either way: the rate enters only through the excess return in $\widehat{SR}_d$, and the DSR of §A.6.2 is 99.5% at $r_f = 1\%$, 98.8% at 2%, 98.1% at about 2.7%, and 97.6% at 3%.

Not to be confused with the margin financing rate discussed elsewhere (3.5%): that is the cost of borrowed capital, this is the term subtracted in the numerator of the Sharpe ratio. They are two different quantities that happen to live in the same section of a backtest.

A.4.4 Number of observations, $n = 2011$

Daily equity returns, sampled on **trading days** between 2017-05-10 and 2025-05-10. There are 2,012 trading days in the window, and the first observation produces no return.

LEAN's series instead contains 911 observations with identically zero return (weekends and holidays): they are not observations, they are padding, and the series on trading days is the statistically correct one.

The result does not depend on the choice, however, and it is worth showing this explicitly, since elsewhere in the report LEAN's sampling convention is documented as a source of confusion. Under dilution by a factor κ , here $\kappa = 2922/2012 = 1.45229$, all the quantities that enter the DSR transform in a coordinated way:

$$\widehat{SR} \rightarrow \frac{\widehat{SR}}{\sqrt{\kappa}}, \quad \sqrt{n-1} \rightarrow \sqrt{\kappa} \sqrt{n-1}, \quad \hat{\gamma}_3 \rightarrow \hat{\gamma}_3 \sqrt{\kappa}, \quad \hat{\gamma}_4 \rightarrow \hat{\gamma}_4 \kappa$$

The numerator $\widehat{SR} \sqrt{n-1}$ is invariant because it is, in essence, total return over total risk. The denominator is invariant up to one term: $\hat{\gamma}_3 \widehat{SR}$ and $\hat{\gamma}_4 \widehat{SR}^2$ are both invariant under these transformations, but the constant -1 inside $(\hat{\gamma}_4 - 1)/4$ is the single piece that does not rescale

with the moments, and it is what produces the 0.92150 against 0.92182 of the table below. Because it multiplies $\widehat{SR}^2$, its effect on the DSR is of order 10^{-5} . And the threshold SR_0 rescales by $1/\sqrt{\kappa}$ as well, because the SE goes as $1/\sqrt{n}$.

	$\widehat{SR}$	n	denominator	z	DSR
trading days	0.08694	2011	0.92150	2.2728	98.85%
calendar days	0.07214	2922	0.92182	2.2727	98.85%

There is therefore no knob to turn: the conclusion is the same whichever convention is adopted. The invariance holds to the precision reported rather than exactly. Besides the constant -1 above, the dilution of the moments is itself an approximation, because the m^2 term in $m_2 = \sum_i r_i^2/n - m^2$ scales with κ^2 and not with κ , and the same applies to the centring terms of the third and fourth moments. Redoing the whole calculation on the calendar series from the raw power sums, with none of these approximations, gives the same figure to within a hundredth of a point.

Corollary. What makes the DSR of this appendix invariant is that its threshold SR_0 is built from the standard error and therefore rescales with the sampling, together with everything else. A statistic whose threshold is a fixed number instead does not survive the change: LEAN's reported Sharpe subtracts an annual risk-free rate that does not dilate with κ , and its PSR compares against the constant $1/\sqrt{252}$ (§A.8). Both are convention-dependent for the same reason, and that is why this appendix uses neither.

Sampling is scheduled one minute before the close, hence **after** the forced liquidation at 15:58: returns are measured with the portfolio already flat. The exception is the seventeen half-day sessions of §7, where the liquidation branch never fires and the sample is taken at 12:59 with the position still open: seventeen observations out of 2,011.

A.4.5 Units

$\widehat{SR}_d$ is computed by us per observation on trading days; σ_{cross} instead derives from QuantConnect's Sharpes, annualized with a different convention. The two scales are not linked by a simple factor, because the risk-free term does not rescale with the rest: §A.8 gives the relation in full. To convert a **dispersion**, however, the additive offset cancels and the correct factor is the multiplicative part alone. That part is $\sqrt{252}/\sqrt{\kappa} = \sqrt{252/\kappa} = 13.173$, namely LEAN's annualizer $\sqrt{252}$ applied to a standard deviation already reduced by $\sqrt{\kappa}$ through the calendar-day sampling. In the main specification the question does not arise at all: the SE is computed directly in daily units on the series we measured, so the result that supports the conclusions does not depend on LEAN's annualization convention.

A.5 Instrumentation

The QuantConnect Free plan offers neither API access nor the Object Store, so the daily return series cannot be exported programmatically. It is not needed: skewness and kurtosis are derived from five scalar accumulators.

```
def record_equity_moment(self):
    if self.is_warming_up:
        return
    v = self.portfolio.total_portfolio_value
    if v <= 0:
        return
    if self.prev_equity is not None and self.prev_equity > 0:
        r = v / self.prev_equity - 1.0
        self.ret_n += 1
        self.ret_s1 += r
        self.ret_s2 += r * r
        self.ret_s3 += r * r * r
        self.ret_s4 += r * r * r * r
    self.prev_equity = v
```

The five values are emitted via `set_runtime_statistic`, a channel that does not consume the 10 kb log quota of the Free plan. Exponential formatting with 12 digits: the sums of fourth powers are of order 10^{-5} and fixed-point formatting would zero them out.

From these, the sample moments:

$$m_1 = \frac{S_1}{n}, \quad m_2 = \frac{S_2}{n} - m_1^2, \quad \hat{\gamma}_3 = \frac{S_3/n - 3m_1S_2/n + 2m_1^3}{m_2^{3/2}}$$
$$\hat{\gamma}_4 = \frac{S_4/n - 4m_1S_3/n + 6m_1^2S_2/n - 3m_1^4}{m_2^2}$$

Cost: **one backtest**, on S4' tight at 0 bps.

A.6 Results

A.6.1 Measured quantities

quantity	value
n	2011
daily mean m	$6.7596 \cdot 10^{-4}$

quantity	value
daily standard deviation s	$6.8622 \cdot 10^{-3}$
$\hat{\gamma}_3$ (skewness)	+2.011
$\hat{\gamma}_4$ (kurtosis, non-excess)	13.703

One quantity is derived rather than measured, and a choice enters with it that does not enter above, the risk-free rate of §A.4.3: the sample Sharpe per observation, $\widehat{SR}_d = (m - r_f/252)/s = 0.08694$.

A.6.2 Main result

With $SE = 0.02234$ and $N = 16$:

$$SR_{0,d} = 0.02234 \times 1.8005 = 0.04022$$

$$z = \frac{(0.08694 - 0.04022)\sqrt{2010}}{\sqrt{1 - 2.011 \cdot 0.08694 + \frac{13.703-1}{4} \cdot 0.08694^2}} = \frac{2.0946}{0.9215} = 2.273$$

$$\boxed{DSR = \Phi(2.273) = \mathbf{98.85\%}}$$

With the exact null value of the SE, $1/\sqrt{n} = 0.02230$, the threshold is 0.04015 and the DSR 98.86%: the plug-in costs a hundredth of a point.

How the figure should be stated. It is a confidence level in the sense of §A.2, equivalently the complement of a one-sided p-value, and not a posterior probability, so the defensible wording is: *under the null hypothesis that all sixteen configurations have a true Sharpe of zero, the observed maximum is incompatible with selection noise at the 5% level ($z = 2.27$, $p = 0.012$, one-sided).* This says the same thing as "DSR = 98.85%" and claims nothing the construction does not support. The level is nominal rather than exact, for the reasons given in §A.10.

A.6.3 Consistency check of the instrumentation

The results of this appendix rest entirely on the five accumulators, and so far nothing verifies their correctness. The check that follows is free and uses only numbers already available; its scope, what it establishes and what it does not, is delimited below.

The compounded return over the whole period is $\prod_i (1 + r_i)$, and its logarithm expands into the power sums already collected:

$$\sum_i \ln(1 + r_i) = S_1 - \frac{S_2}{2} + \frac{S_3}{3} - \frac{S_4}{4} + O(S_5)$$

order	value	relative gap
1	1.359364699	$3.6 \cdot 10^{-2}$
2	1.311556089	$3.7 \cdot 10^{-4}$
3	1.312055857	$1.25 \cdot 10^{-5}$
4	1.312039632	$1.4 \cdot 10^{-7}$

vs. $\ln(1 + 2.71374) = 1.312039455$, derived from the net profit reported by QuantConnect.

What this verifies. Two things. First, the **endpoints**: the first sample starts from $V_0 = 100,000$ and the last coincides with the final equity that produces the reported +271.374%. Second, and more important, the **mutual consistency of the four accumulators**: if S_3 or S_4 contained an error, the corresponding term would not close the residual of the previous one. That the agreement improves by two orders of magnitude in going from the third to the fourth term (a factor above ninety) instead of leveling off rules out systematic errors in the higher moments, which are exactly the ones on which the DSR denominator rests.

What this does not verify. The check can say nothing about *which* instants were sampled, because the product telescopes. Every r_i is built on `prev_equity`, that is, on the last recorded value and not on the previous day's, so

$$\prod_i (1 + r_i) = \frac{V_{\text{last}}}{V_{\text{first}}}$$

for **any** subset of instants. If a day were skipped, `prev_equity` would not update and the next return would cover two sessions: the product would remain identical, n would drop to 2010 and the check would pass anyway. The same holds if sampling occurred at 15:00 rather than at 15:59.

The check on n has to be done separately, and it is needed, because $\sqrt{n-1}$ enters the z of §A.6.2 directly. An independent count of NYSE sessions between 2017-05-10 and 2025-05-10 gives **2,012** (the last is May 9, since the 10th falls on a Saturday), hence 2,011 returns: this coincides with `ret_n`.

One further gap, closed elsewhere. Everything above compares the accumulators with each other and with the endpoints; all of it is internal to the five numbers, and none of it confronts them with a quantity computed by anyone else. §A.8 does exactly that: it reproduces LEAN's annualized volatility, its Sharpe and its PSR from s , $\hat{\gamma}_3$ and $\hat{\gamma}_4$, against figures the platform computed from its own series. Had the sampling taken different instants, those moments would not be ours diluted by κ and none of the three would land. The constraint is coarse, the printed volatility carries two significant figures and the half point on the PSR is already

absorbed by the band on the platform's risk-free rate, but it is the only external check on the moments in this appendix, and it bears on precisely the point the telescoping argument leaves open.

A.7 Sensitivity to the number of trials

N	Gumbel bracket	$SR_{0,d}$	DSR
16	1.8005	0.04022	98.85%
50	2.2763	0.05086	96.04%
65	2.3751	0.05306	95.03%
100	2.5306	0.05654	93.04%
500	3.0525	0.06820	81.91%
1,000	3.2551	0.07272	75.54%

The exact 95% threshold falls at $N = 65.5$. The conclusion therefore stays above 95% up to $N \approx 65$ trials, vs. the actual 16: a margin of about a factor of four on the number of configurations it would have been legitimate to explore. With the exact null SE the threshold moves to $N \approx 66$.

It is a solid margin but not an unlimited one, and this is the correct formulation of the result.

For comparison, the Bailey and López de Prado specification. Bringing $\sigma_{\text{cross}} = 0.06795$ into daily units, that is, dividing it by 13.173 (§A.4.5), the DSR is 99.99% at $N = 16$, 99.94% at $N = 10^4$ and 99.65% at $N = 10^9$: no number of trials brings the conclusion down. This total insensitivity is not robustness; it is the diagnostic symptom that the cross-sectional dispersion is too small to be a sensible null dispersion when the trials are as correlated as these. The number is reported for completeness, not offered as a result.

A.8 How LEAN annualizes, and what its printed figures mean

Three statistics of the same run are quoted from QuantConnect elsewhere in this report, the annualized volatility, the Sharpe ratio and the PSR, and all three depend on a convention of the platform rather than on the strategy. The convention is visible in the source, and the moments measured above are enough to check that it has been read correctly.

LEAN samples the performance series once per calendar night, not once per session (`BaseResultsHandler.Sample`, whose value is $(E_t - E_{t-1})/E_{t-1}$, so a night with no trading

contributes an exact zero), and annualizes that series with 252 (`PortfolioStatistics: AnnualVariance(listPerformance, tradingDaysPerYear)`).

The series it works on is therefore ours **diluted**: 2,011 real returns spread over 2,922 observations, the other 911 being zero, so $m_{\text{cal}} = m/\kappa$ and $s_{\text{cal}} = s/\sqrt{\kappa}$ with $\kappa = 2922/2012 = 1.45229$. The risk-free rate is subtracted as an annual figure and does not rescale with κ , so the reported Sharpe is

$$SR_{QC} = \frac{(1 + m/\kappa)^{252} - 1 - r_f}{\sqrt{252/\kappa} s}$$

The numerator is compounded and not linear, because `Statistics.AnnualPerformance` raises the mean plus one to the power `tradingDaysPerYear`; see below on how that was established.

Written with a linear numerator the same expression becomes $\sqrt{252/\kappa} (m - \kappa r_f/252)/s$, which is algebraically transparent and shows where each κ acts: the first is LEAN's annualizer $\sqrt{252}$ applied to a series that has $252\kappa = 366$ observations per year, the second is a risk-free rate charged 366 times a year at a daily rate calibrated on 252, that is, an effective $\kappa r_f = 3.92\%$. It is not, however, what LEAN computes, and two warnings are in order for anyone checking the arithmetic. At $r_f = 2.70\%$ the linear form returns 0.999 against the 1.075 printed. And at the $r_f = 2\%$ of this report it returns 1.076, which lands on the printed value through the cancellation of two errors and must not be taken as a reproduction. The compounded form at 2.70% gives 1.078, and that is the reproduction.

The check is that this reproduces what QuantConnect prints, from our moments and nothing else:

quantity	from our moments	reported by QC
annualized volatility, $\sqrt{252} s/\sqrt{\kappa}$	0.0904	0.09
Sharpe, LEAN's convention, $r_f = 2.70\%$	1.078	1.075
Probabilistic Sharpe Ratio	49.8%	49.490%
(for contrast) Sharpe on trading days, $\widehat{SR}_d\sqrt{252}$, linear	1.38	—

The first line has no free parameter: κ comes from the calendar and s from the measurement. The second and third add only the platform's risk-free rate, and the third runs LEAN's own routine (`Statistics.ProbabilisticSharpeRatio`) on the same diluted moments, landing within half a point of the printed value; together they settle the reading of the convention. The fourth line is not a reproduction but a comparison: the same run annualized on the days it actually traded, which is what the printed figure understates, here and for every QC Sharpe

in this report. It uses the linear annualization, so it is not on the convention of the lines above; under LEAN's compounded form the same series on trading days would give 1.52. Either number makes the point. The QC figures are quoted as printed throughout, for comparability with the paper.

Two details are needed to reproduce the PSR, and they matter for §7. LEAN sets its `benchmarkSharpeRatio` to $1/\sqrt{252}$, that is, it measures the PSR against a **Sharpe of 1** rather than against zero, so the number it prints is not the probability that the true Sharpe is positive. And the risk-free rate it uses is not a convention but its interest rate model: the Sharpe inverted on five independent runs of this report puts it between 2.69% and 2.72%, the short rate over this window. That inversion also settles how the numerator is annualized. `Statistics.AnnualPerformance` raises the mean plus one to the power `tradingDaysPerYear`, which compounds it, while the summary comment above the method describes the linear form instead; under the compounding form the five runs agree on the rate, under the linear one they spread over more than a point, so the runs and the source say the same thing.

This says nothing about the strategy, and it changes no result of this report: it fixes how the platform statistics quoted elsewhere should be interpreted. The risk-free rate the report adopts for its own Deflated Sharpe Ratio, $r_f = 2\%$, remains a convention declared in advance (§A.4.3), not a quantity taken from the platform.

A.9 Comment

A.9.1 Positive skewness works in favor, the tails barely matter

The denominator is 0.9215, and it decomposes as follows:

$$\underbrace{1 - \hat{\gamma}_3 \widehat{SR}_d}_{-0.1748} + \underbrace{\frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}_d^2}_{+0.0240} = 0.8492 \quad \Rightarrow \quad \sqrt{0.8492} = 0.9215$$

Without skewness the denominator would be 1.012, that is, the DSR would be lower. The contribution of skewness is **seven times** that of kurtosis, even though the kurtosis is 13.7 (excess: 10.7): the kurtosis term enters squared in $\widehat{SR}_d$, which is 0.087, and vanishes numerically. The ratio is a consequence of the daily frequency and not a general hierarchy: skewness multiplies $\widehat{SR}_d$ and kurtosis multiplies $\widehat{SR}_d^2$, so on annual Sharpes, where $\widehat{SR} \approx 1$, the order reverses and the kurtosis term is the one that dominates.

The common intuition is that fat tails penalize the Sharpe ratio as an indicator; here the penalty exists but is second order, while positive skewness dominates.

A.9.2 The skewness is consistent with the rest of the picture

$\hat{\gamma}_3 = +2.01$ means many small losses and few large wins: the canonical profile of a momentum strategy. It is consistent with the other statistics reported for S4', a win rate of 40% and a high profit/loss ratio, which are independent measures and tell the same story. This is also a **desirable** property that the Sharpe ratio alone does not show: at equal Sharpe, a distribution with positive skewness is preferable.

A.10 Limitations

The DSR corrects for selection within the family, not for the choice of the family. The sixteen configurations are minor variants of the same strategy, on the same instrument, over the same window. The test answers: *given this family, is the best distinguishable from the family's noise?* It does not answer: *does intraday momentum on SPY have edge?* That selection took place upstream, at the moment the family was published because it worked, and no correction applied downstream can recover it.

The threshold depends on the data only through a plug-in. In a Wald test the null value θ_0 is a constant fixed in advance. Under H_0 the standard error of a zero Sharpe is $1/\sqrt{n}$, a constant, and so would be the threshold SR_0 . The main specification of §A.4.2 plugs the observed $\widehat{SR}_d$ into the SE instead, which ties SR_0 to the same $\widehat{SR}_d$ that stands in the numerator, so that the variance of $\widehat{SR}_d - SR_0$ is not exactly the SE^2 the denominator uses. The tie is immaterial here: with the constant null value, 0.02230 instead of 0.02234, the DSR is 98.86% instead of 98.85%. The level remains nominal because of the approximations below, not because of the threshold.

Independence of observations. The factor $\sqrt{n-1}$ assumes independent daily returns. Residual autocorrelation would reduce the effective sample size and therefore the DSR. It was not tested. The cost is not quite zero, since the return series itself is not available (§A.5), but it is close to it: one further accumulator, $S_{11} = \sum_i r_i r_{i-1}$, yields $\hat{\rho}_1 = (S_{11}/n - m^2)/m_2$ alongside the five already collected, and two more extend this to a Ljung-Box on the first three lags. One backtest.

Normality of the maximum. The Gumbel approximation for SR_0 assumes that the Sharpes under H_0 are Gaussian. With $n = 2011$ observations this is reasonable, but the marked skewness of the returns weakens it. The effect goes in the unfavorable direction: positive skewness in the distribution of sample Sharpes thickens the right tail and therefore raises SR_0 . It is of order $1/\sqrt{n}$, however, hence negligible with $n = 2011$.

Single window. The DSR is not an out-of-sample validation: it says that the result is not an artifact of the *search*, not that it holds in different sub-windows of time. That is a separate question, and it remains open; see §8.

A.11 Conclusions

1. The Sharpe of 1.075 of S4' tight is **not an artifact of the selection among sixteen configurations**: DSR = 98.85% under the conservative specification, 98.86% with the exact null standard error. The test is against the selection threshold SR_0 , under the null that all sixteen configurations have a true Sharpe of zero: it says that S4' is not selection noise, not that it is better than S4, from which it differs by +0.033 in log return over eight years (§6).
2. The result holds up to **about 65 trials** vs. the actual 16, a margin of a factor of four on the parameter a critique could focus on. The standard Bailey and López de Prado specification would give 99.99% and insensitivity to any N , but that insensitivity is an artifact of the correlation between the trials and should not be claimed.
3. The same moments settle how the platform's printed figures should be read (§A.8): LEAN samples the equity series on calendar days and annualizes it with 252, and its Probabilistic Sharpe Ratio is measured against a benchmark Sharpe of 1 rather than against zero. Both reproduce from our moments. This changes no result here; it is what allows §7 to say which platform figures can be cited and which cannot.
4. The return profile of S4' has **marked positive skewness** (+2.01), consistent with the rest of the statistics and favorable in a way the Sharpe ratio does not capture.
5. Two distinct things remain uncovered: the upstream selection of the strategy family, and temporal stability. The second is outside the scope of this report and was not addressed; the first is a structural limit worth noting, not resolvable with these data.
6. The choice to compute the DSR from the raw moments (§A.5) rather than from the PSR in QuantConnect's interface turns out to be necessary, not merely prudent: that statistic, recomputed today on unchanged code, differs by 39 to 46 percentage points from the value preserved in the result files, on all five published strategies (§A.2.1), because the platform has changed its definition in between (§7). A figure whose definition changes under the same name cannot serve as a basis for comparison across dates, and neither version tests against a Sharpe of zero.

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The paper and the repository under analysis are cited in the text as "the paper" and "the repository": [blackswan-quants/intraday-momentum](#) on GitHub, the QuantConnect/LEAN implementation with its accompanying paper and README.